Pattern Recognition
Assignment #1

Diabetes Risk Prediction

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Introduction

This Lab Assignment Focuses on Machine Learning Concepts in multiclass classification.
It envolve the implementation of the following algorithms:

- K-Nearest Neighbors (KNN)
- Softmax Regression
- Neural Networks

The Challenge is to Balance the dataset and to achieve the best possible performance in predicting diabetes risk.

The report is structured as follows:

- Chapter 2: Data Preprocessing
- Chapter 3: K-Nearest Neighbors (KNN)
- Chapter 4: Softmax Regression
- Chapter 5: Neural Networks
- Chapter 6: Evaluation Metrics

Data Preprocessing

2.1 Overview

In this chapter, we will discuss the steps taken to preprocess the diabetes dataset.
The sequence of steps was as follows:

1. Duplicates Removal
2. Data Correlation Analysis
3. Feature Selection
4. Data Splitting
5. Data Balancing
6. Data Normalization

2.2 Data Correlation Analysis

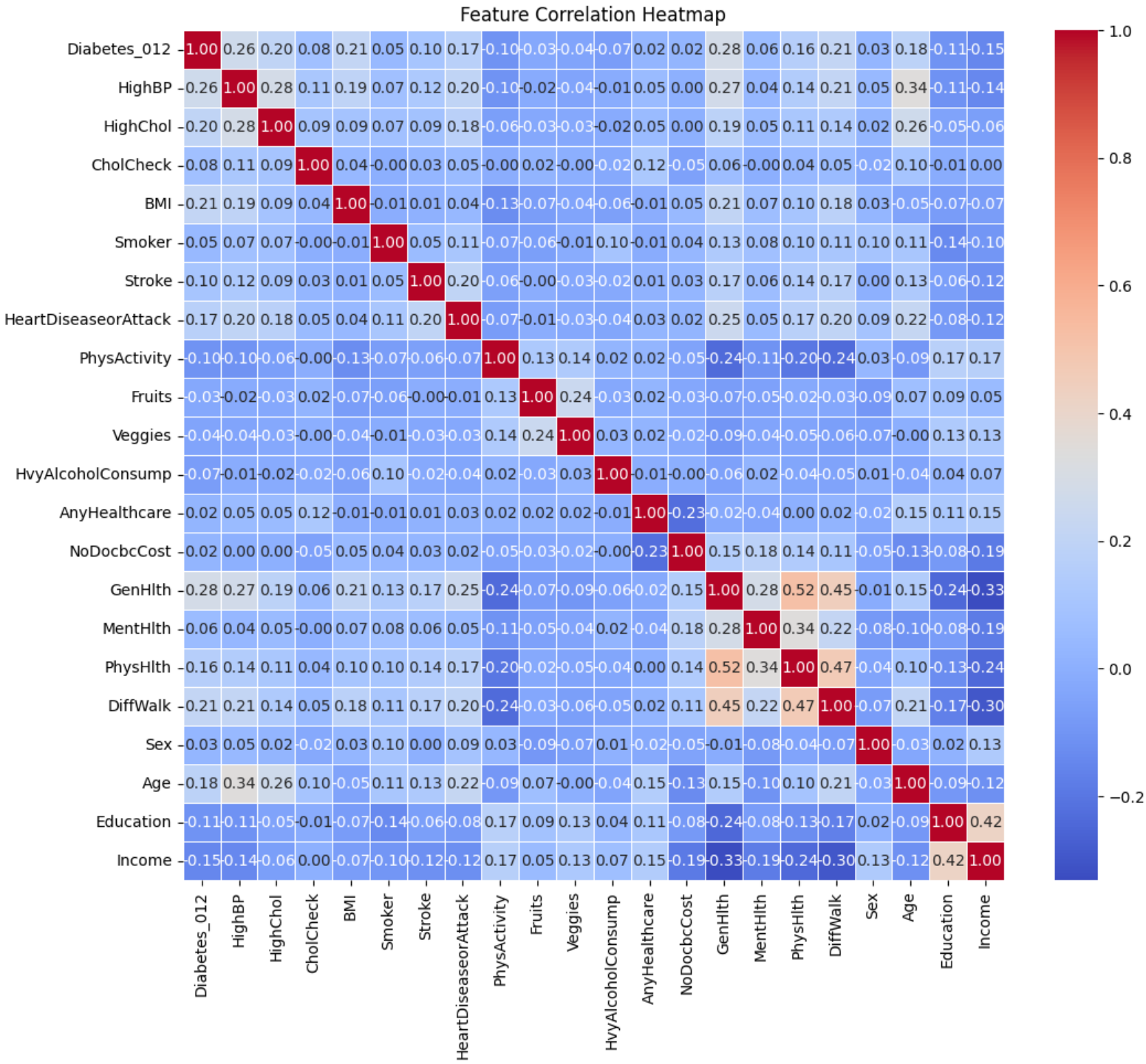

Figure 2.1: Correlation Heatmap of the Diabetes Dataset

2.3 Feature Selection

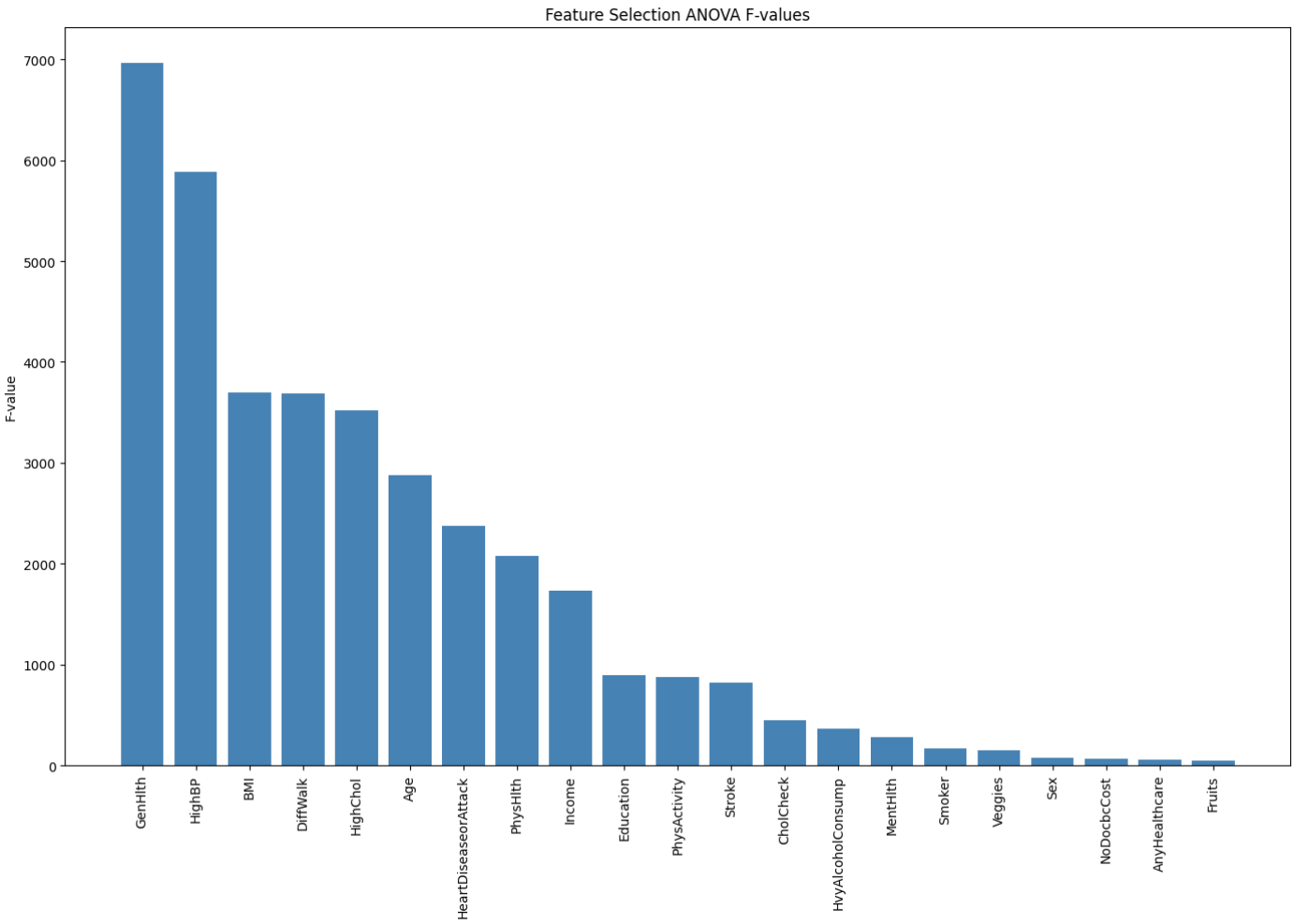

Figure 2.2: Feature Importance Scores from Random Forest Classifier

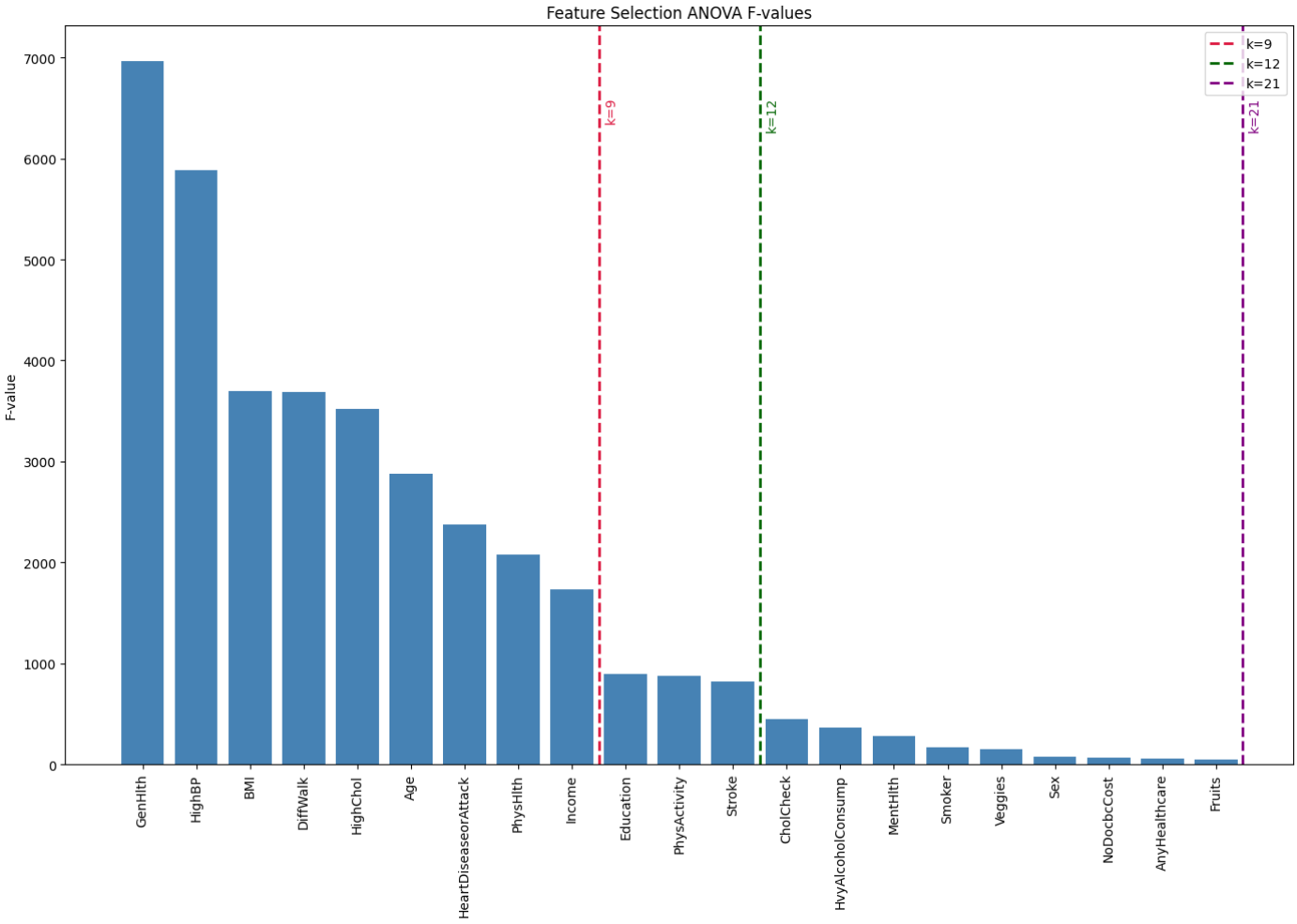

Figure 2.3: Features Selected Based on Feature Importance Scores

The values of k were selected using an elbow-point interpretation of the ranked feature-importance curve. The main bend appears around $k = 9$, after which score gains become more gradual. We then added $k = 12$ as a post-elbow sensitivity check and $k = 21$ as a near-plateau reference to measure the effect of keeping a wider feature set.

Table 2.1: Rationale for Choosing Feature Subset Sizes

Value of k	**Selection Goal**	**Why It Was Selected**
$k = 9$	Elbow point	Chosen as the primary cutoff where the curve changes from steep to flatter growth, capturing the strongest contributors before diminishing returns dominate.
$k = 12$	Post-elbow check	Adds a small number of features after the elbow to test whether modest extra complexity provides meaningful performance or stability gains over $k = 9$.
$k = 21$	Near-plateau reference	Represents the Base Dataset after applying the balancing on it.

2.4 Data Splitting

The dataset was split into :

- Training Set (70%)
- Validation Set (10%)
- Test Set (20%)

The Use of `stratify` parameter in the `train_test_split` function ensured that the class distribution is preserved across the training, validation, and test sets.

2.5 Data Balancing

Data balancing was performed in two ways:

- SMOTE (Synthetic Minority Over-sampling Technique)
- Random Under-sampling

2.5.1 SMOTE

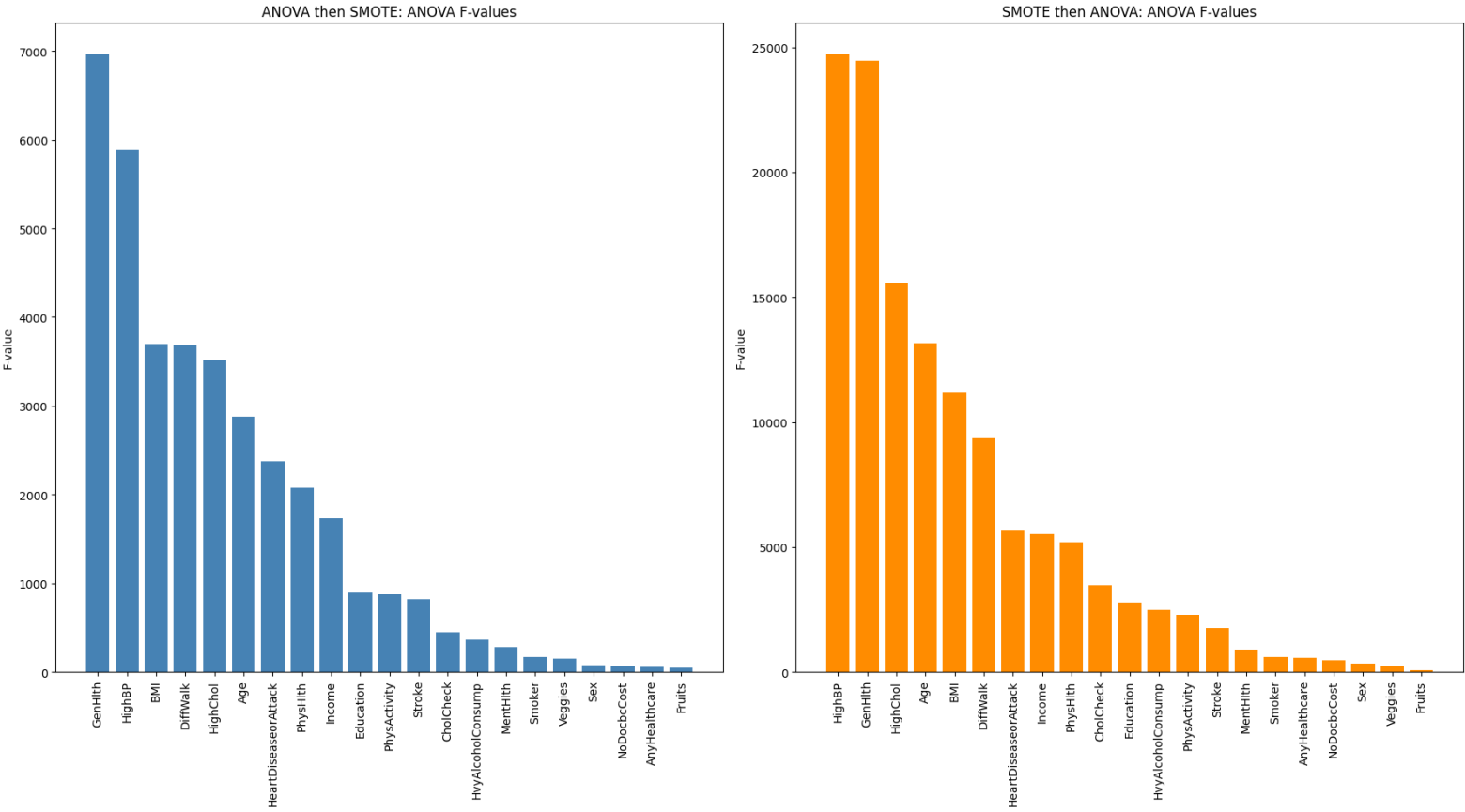

Figure 2.4: Comparison of ANOVA F1-Values before and after balancing

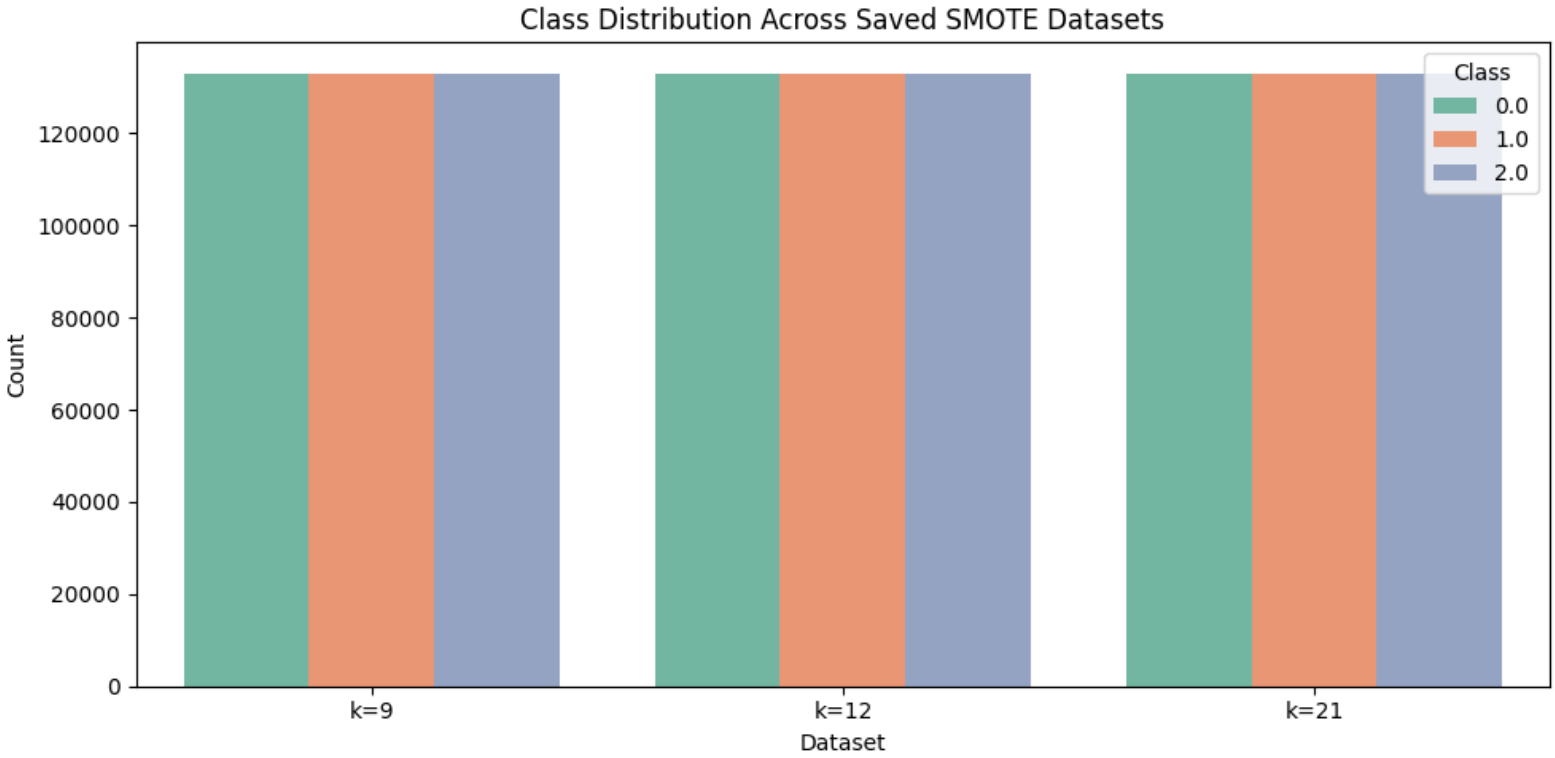

Figure 2.5: SMOTE Class Distribution

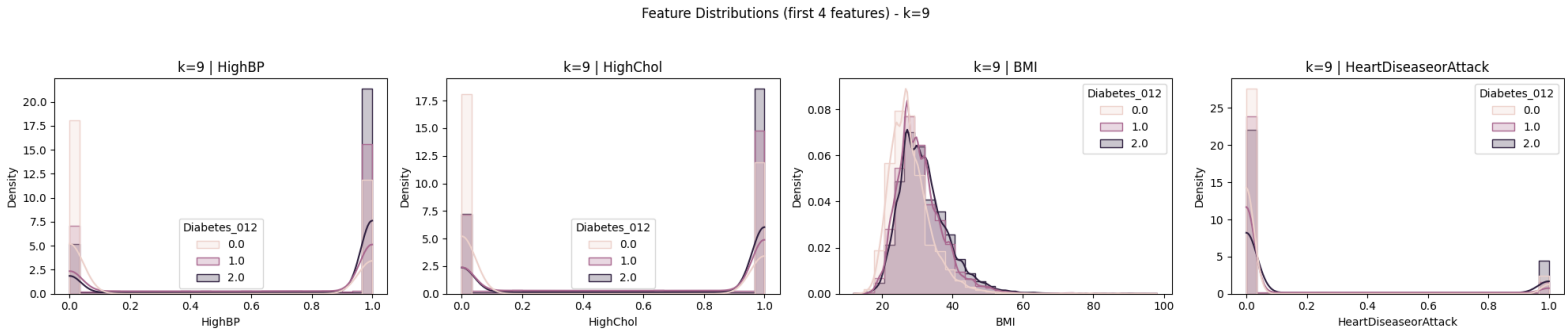

Figure 2.6: Feature Distribution for $k = 9$ after SMOTE Balancing

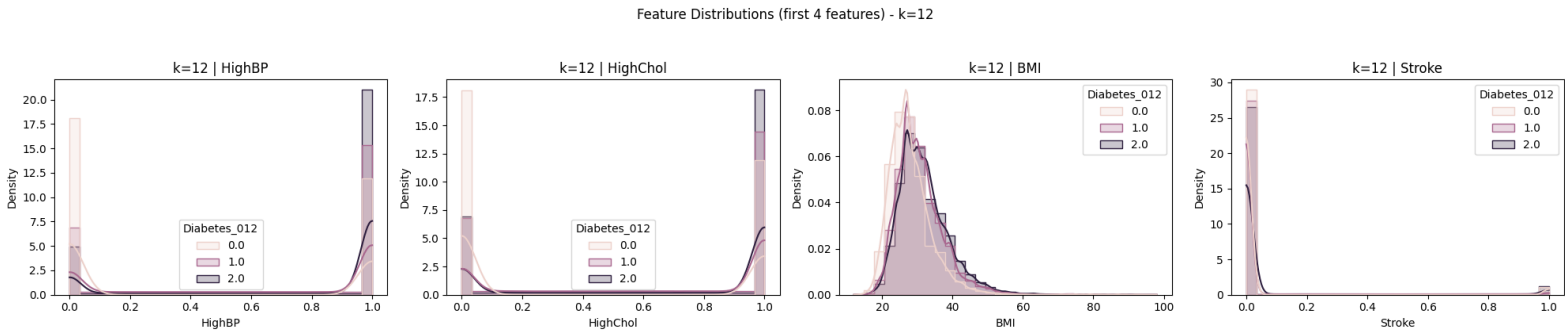

Figure 2.7: Feature Distribution for $k = 12$ after SMOTE Balancing

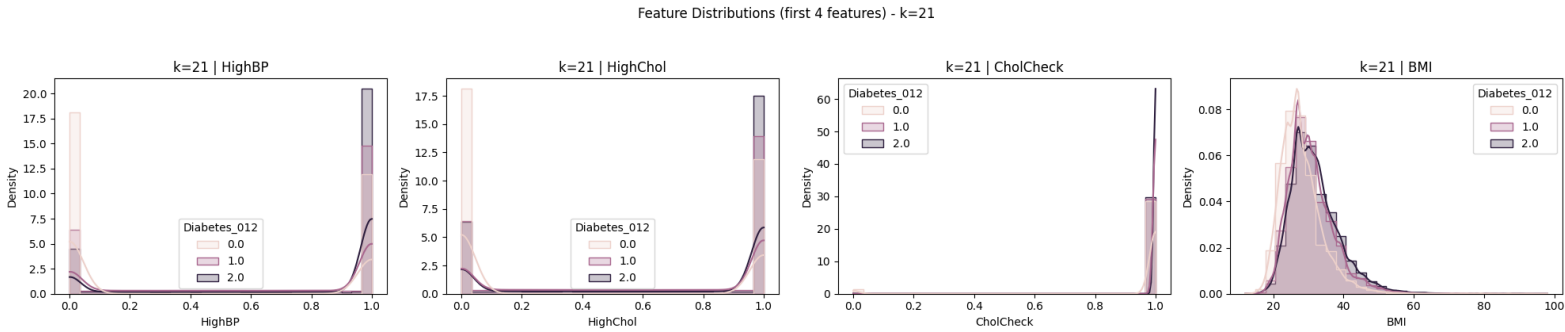

Figure 2.8: Feature Distribution for $k = 21$ after SMOTE Balancing

2.5.2 Random Under-sampling

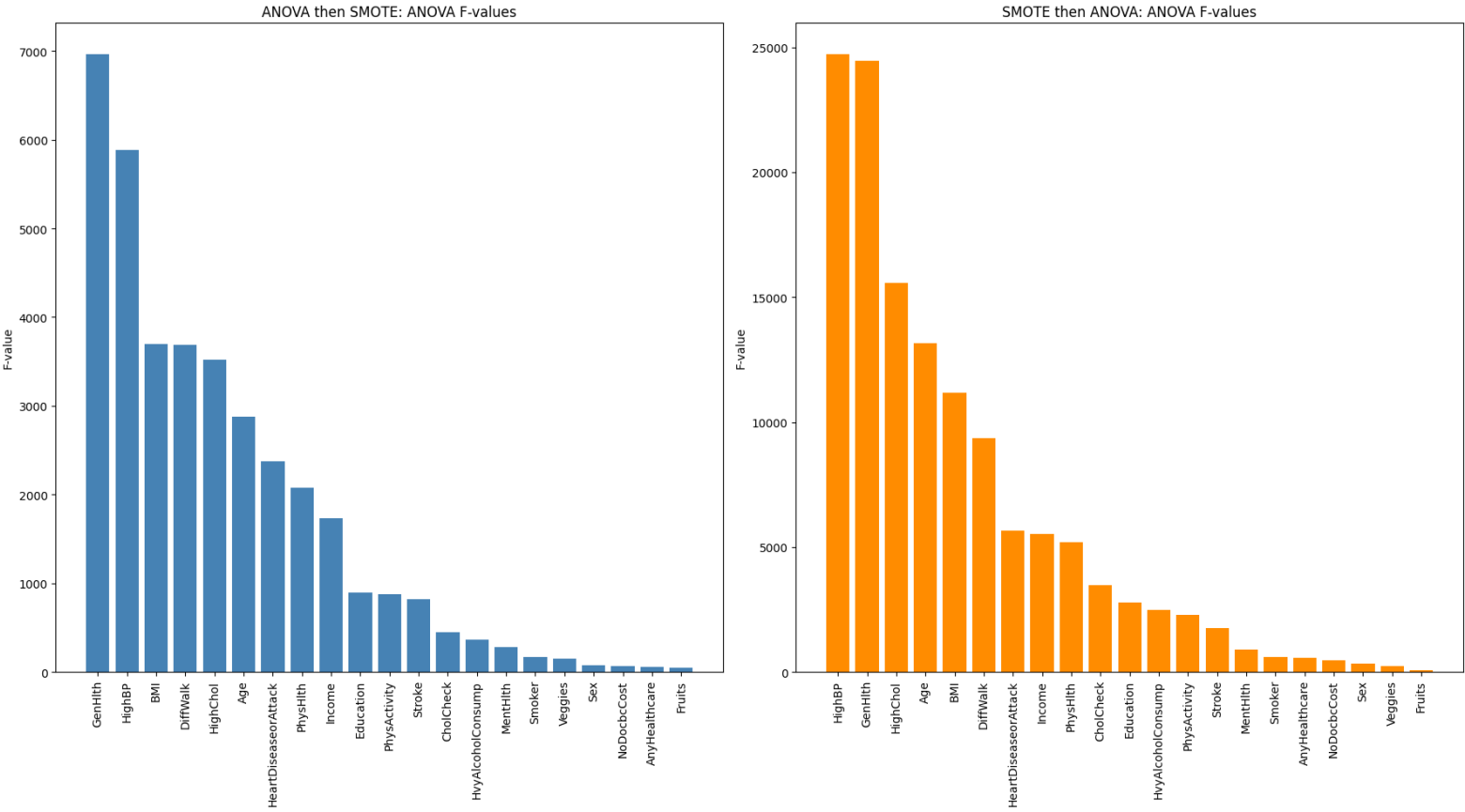

Figure 2.9: Comparison of ANOVA F1-Values before and after balancing

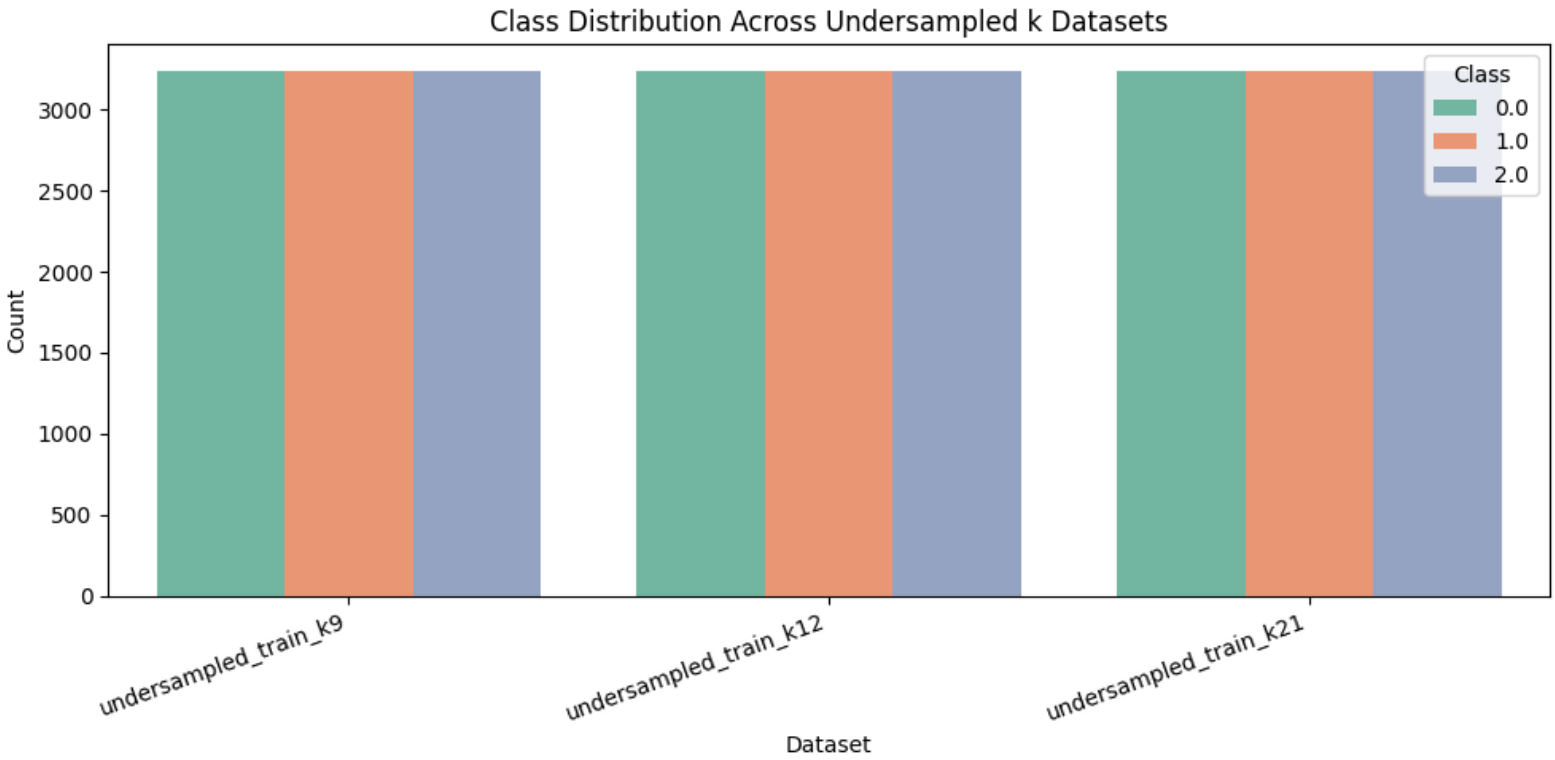

Figure 2.10: Random Under-sampling Class Distribution

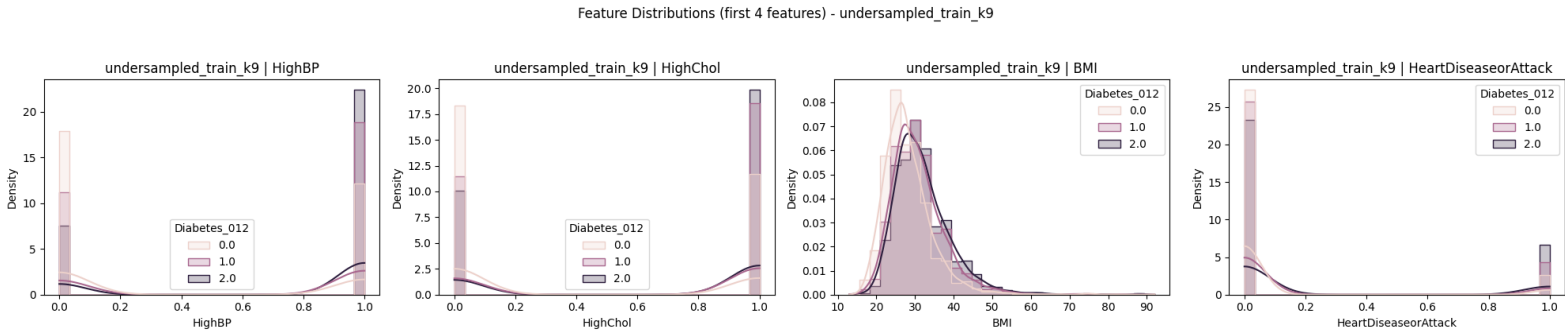

Figure 2.11: Feature Distribution for $k = 9$ after Random Under-sampling

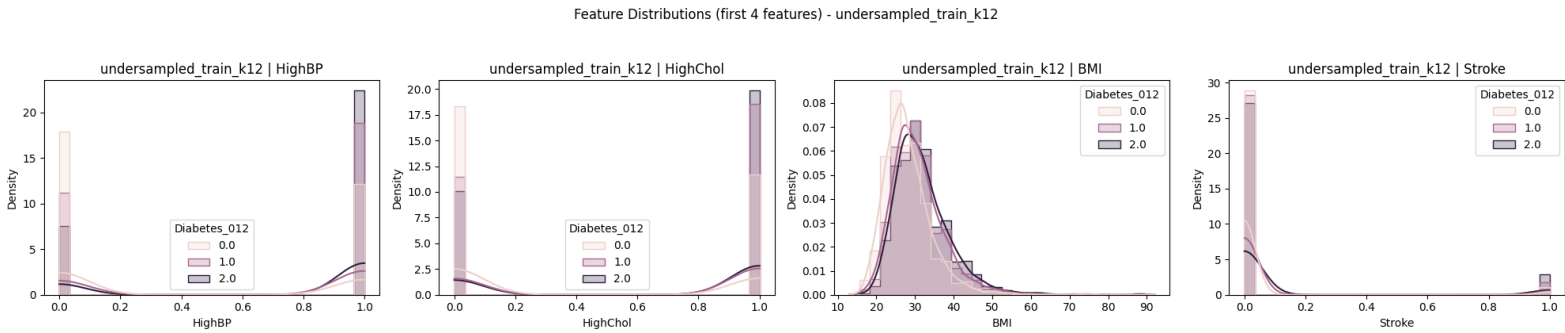

Figure 2.12: Feature Distribution for $k = 12$ after Random Under-sampling

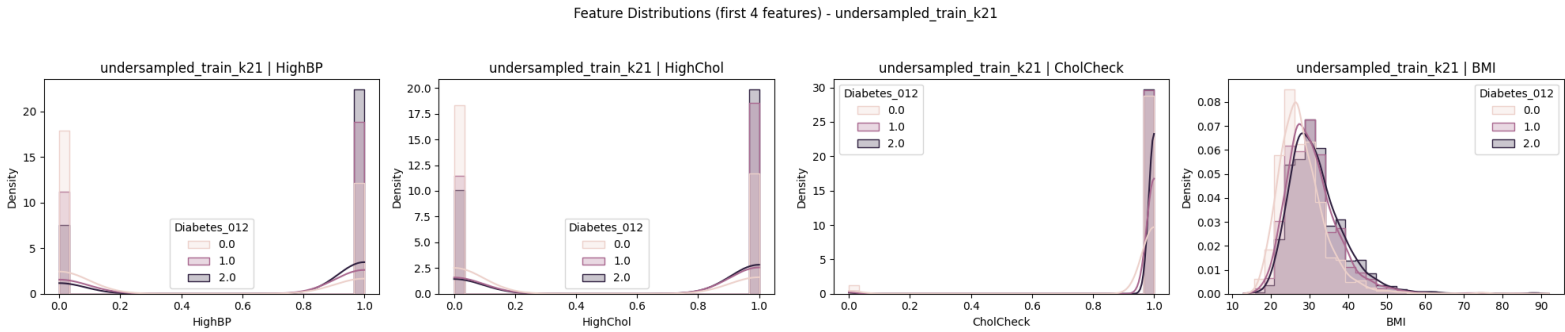

Figure 2.13: Feature Distribution for $k = 21$ after Random Under-sampling

K-Nearest Neighbors (KNN)

3.1 Overview

In this chapter, we will discuss the training of the K-Nearest Neighbors (KNN) algorithm for diabetes risk prediction. We will cover the following topics:

- Evaluation Metrics
- Confusion Matrix

3.2 Evaluation Metrics

Table 3.1: KNN Performance Metrics Across Dataset Variants

Model Variant	**Accuracy**	**F1 Micro**	**F1 Macro**	**F1 Weighted**
Baseline	0.8219	0.8219	0.4016	0.7993
Oversampling ($k = 9$)	0.7298	0.7298	0.4276	0.7579
Oversampling ($k = 12$)	0.7302	0.7302	0.4315	0.7592
Undersampling ($k = 9$)	0.5870	0.5870	0.4020	0.6714
Undersampling ($k = 12$)	0.5908	0.5908	0.4017	0.6735

3.3 Confusion Matrix

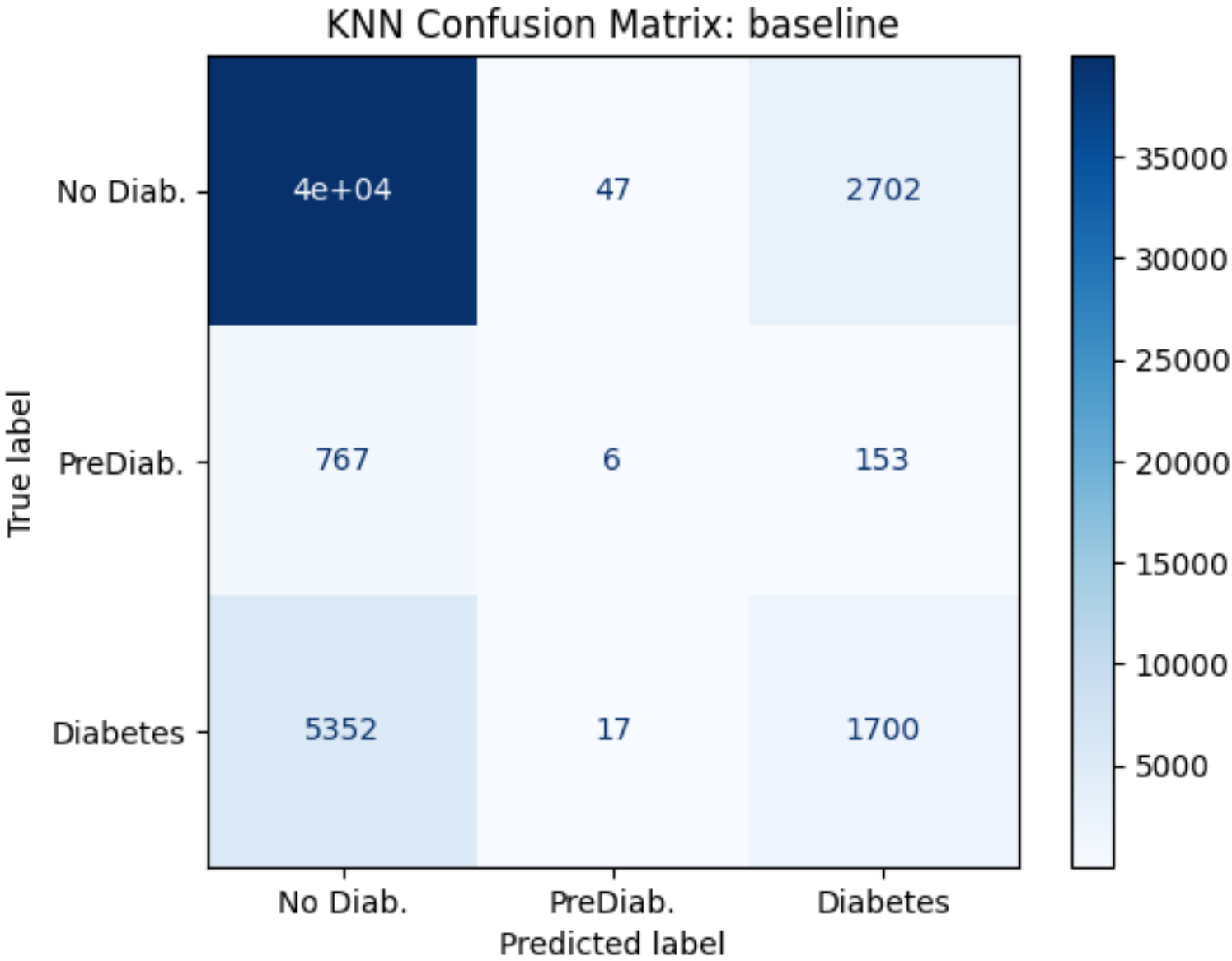

Figure 3.1: Baseline KNN Confusion Matrix

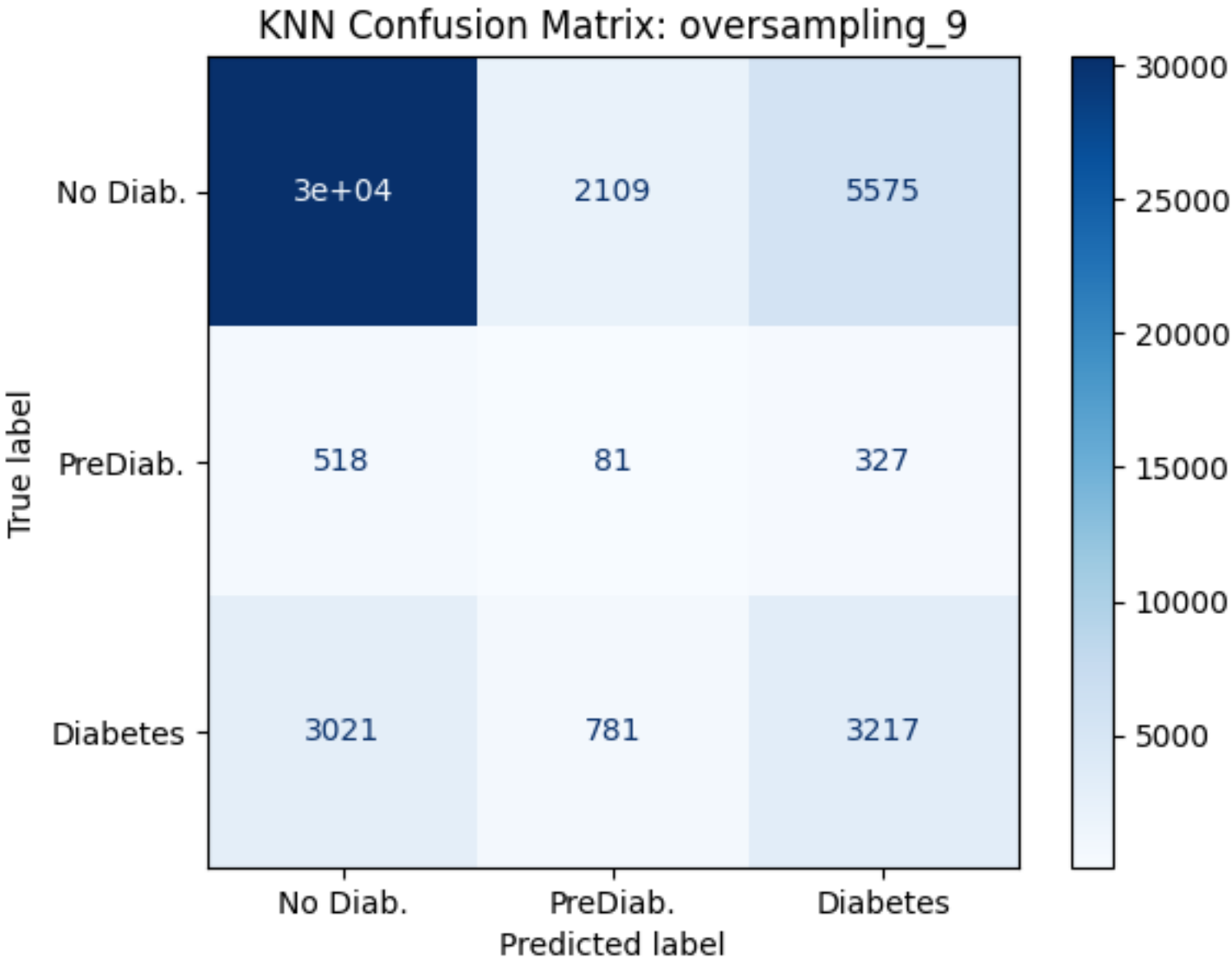

Figure 3.2: SMOTE K=9 KNN Confusion Matrix

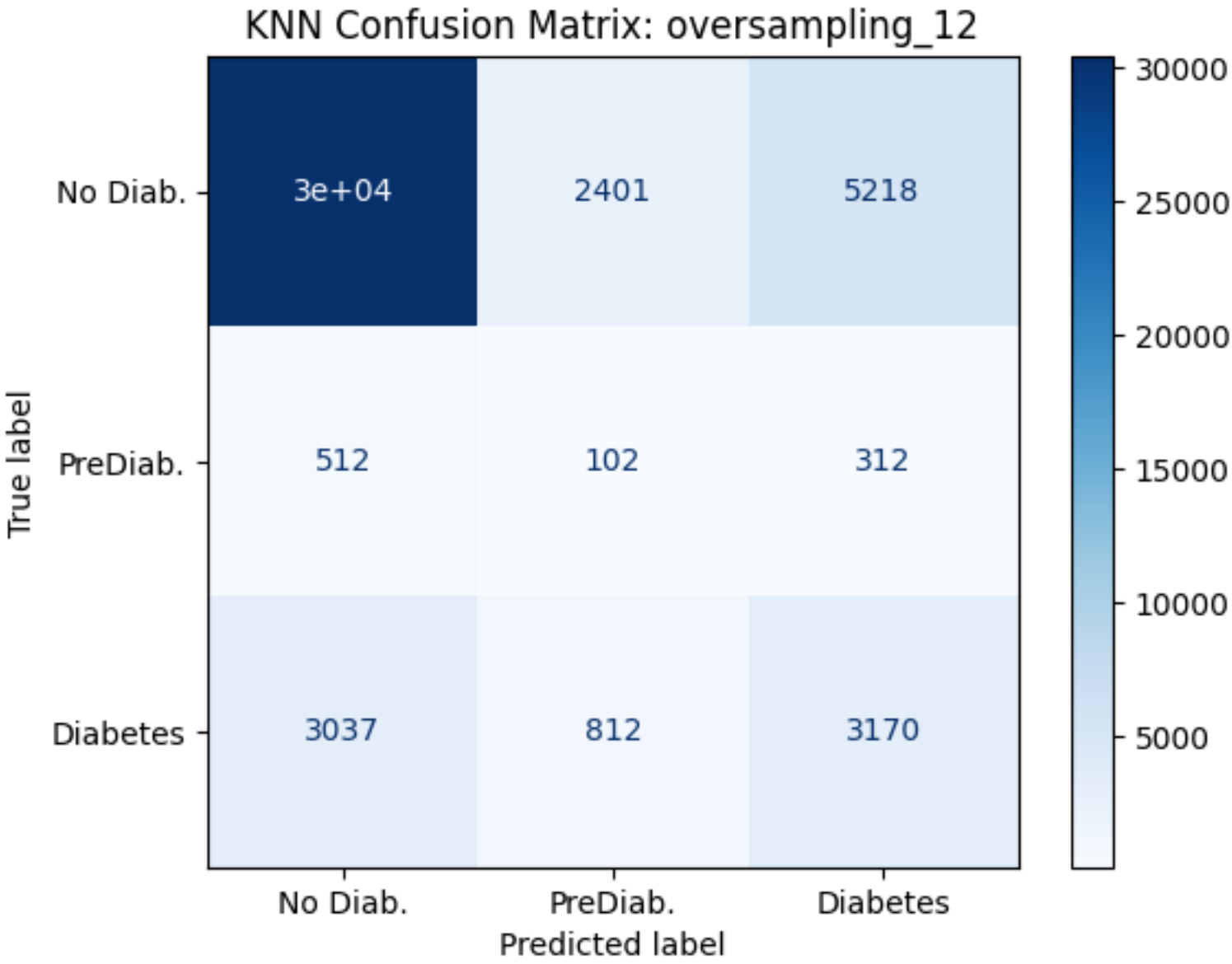

Figure 3.3: SMOTE K=12 KNN Confusion Matrix

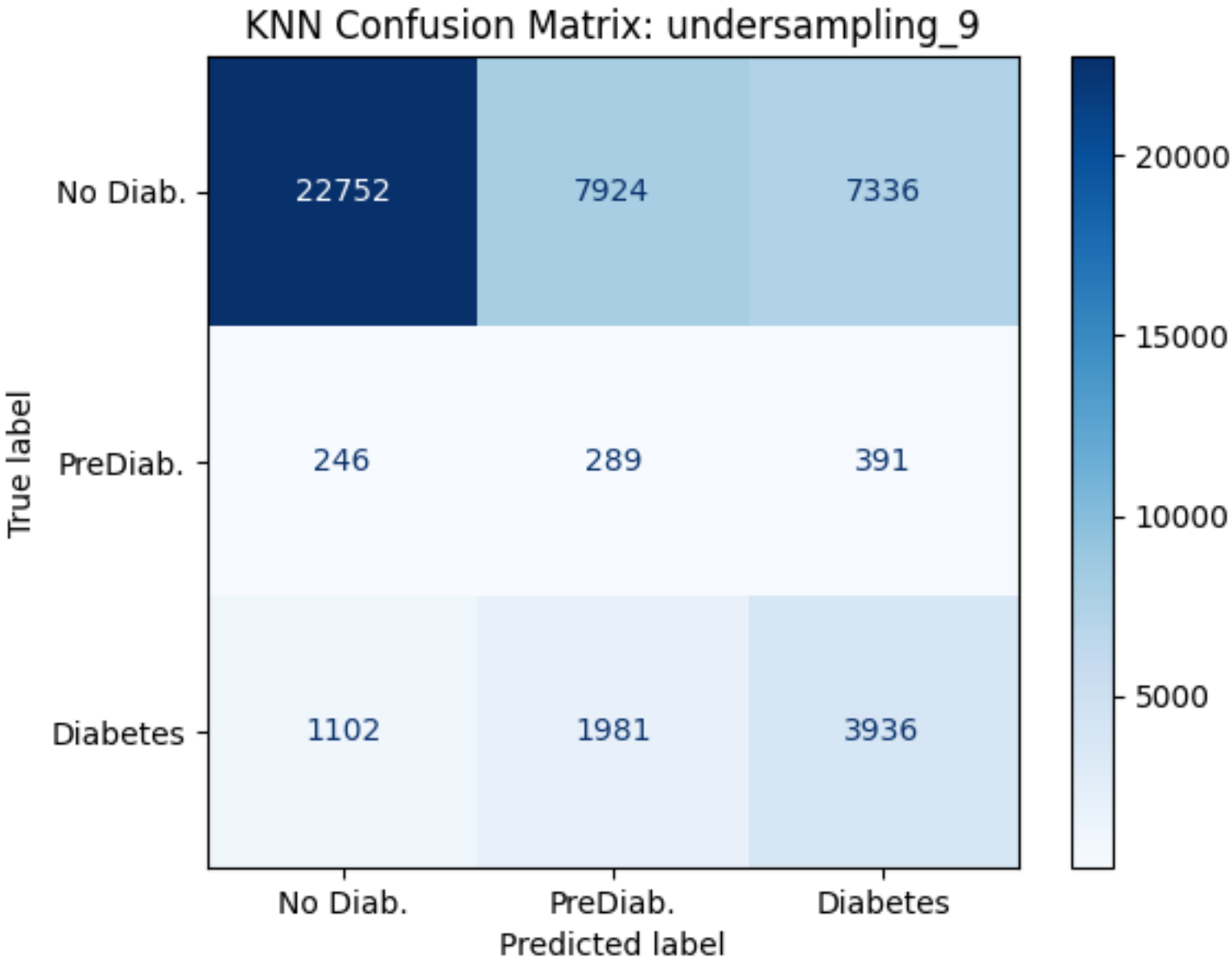

Figure 3.4: Random Under-sampling K=9 KNN Confusion Matrix

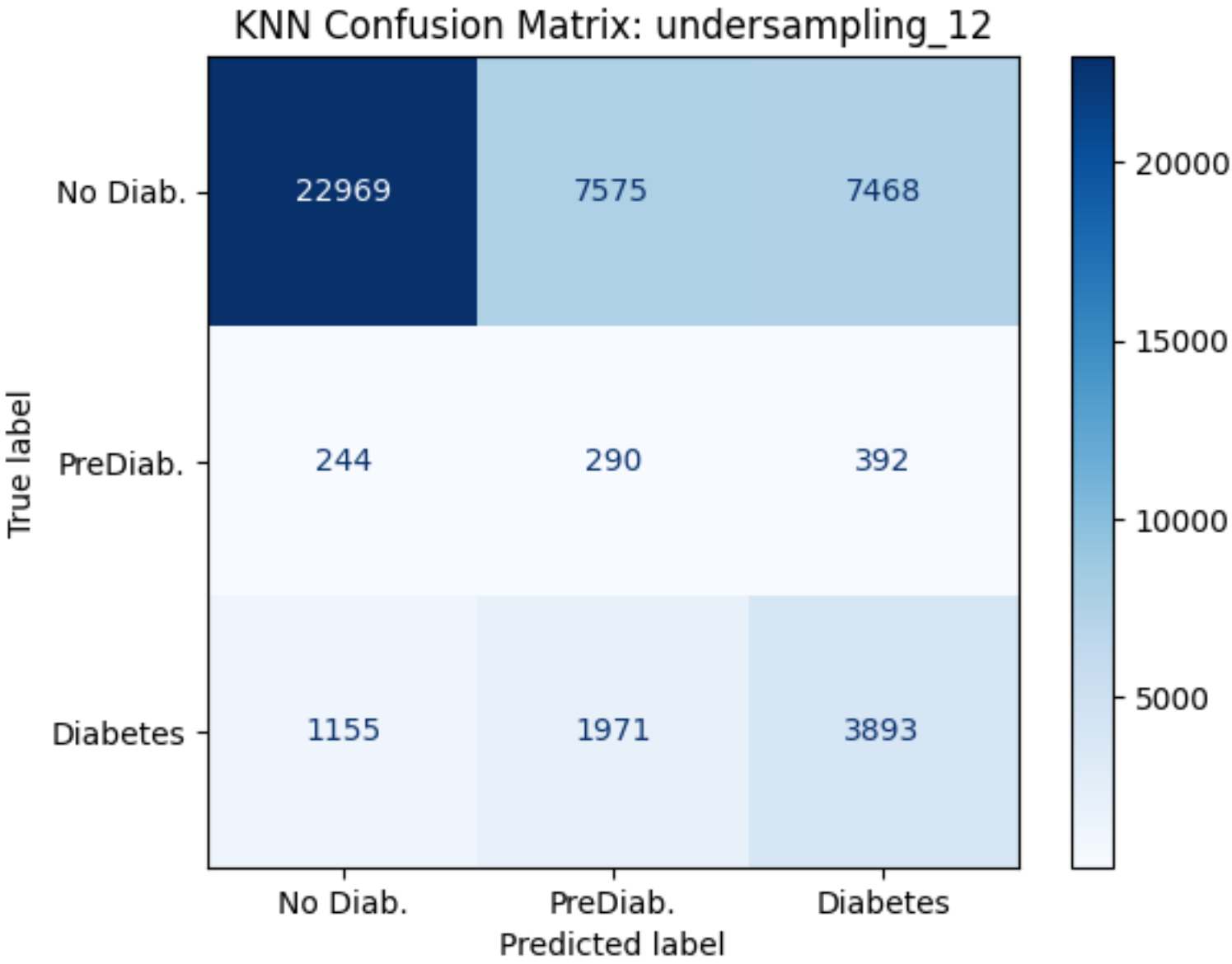

Figure 3.5: Random Under-sampling K=12 KNN Confusion Matrix

Softmax

4.1 Overview

In this chapter, we will discuss the training of the Softmax Regression algorithm for diabetes risk prediction.

4.2 Evaluation Metrics

Table 4.1: Softmax Performance Metrics Across Dataset Variants

Dataset	**Backend**	**Best C**	**Test Accuracy**	**Recall Macro**	**Recall Weighted**	**F1 Micro**	**F1 Macro**	**F1 Weighted**
Baseline	gpu	10.00	0.833540	0.387032	0.833540	0.833540	0.395527	0.793064
Oversampling ($k = 9$)	gpu	0.01	0.630633	0.510483	0.630633	0.630633	0.424164	0.703156
Undersampling ($k = 9$)	gpu	3.00	0.628022	0.510808	0.628022	0.628022	0.423739	0.701904
Undersampling ($k = 12$)	gpu	10.00	0.626325	0.507914	0.626325	0.626325	0.422190	0.700986
Oversampling ($k = 12$)	gpu	0.01	0.628870	0.505213	0.628870	0.628870	0.422111	0.701823
Undersampling ($k = 21$)	gpu	0.01	0.624366	0.511308	0.624366	0.624366	0.421937	0.699614
Oversampling ($k = 21$)	gpu	0.01	0.624976	0.510879	0.624976	0.624976	0.421735	0.699790

4.3 Confusion Matrix

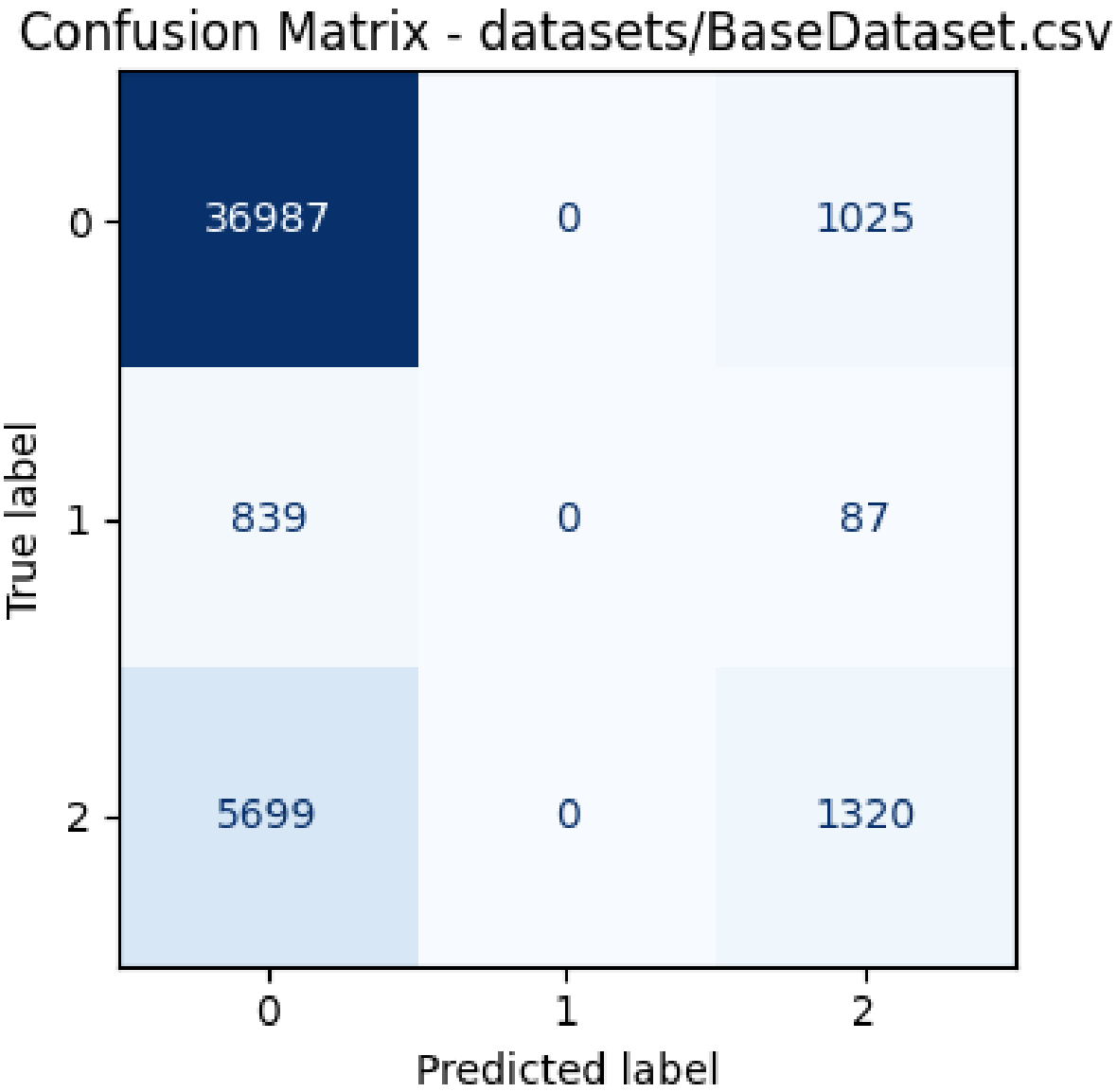

Figure 4.1: Baseline Softmax Confusion Matrix

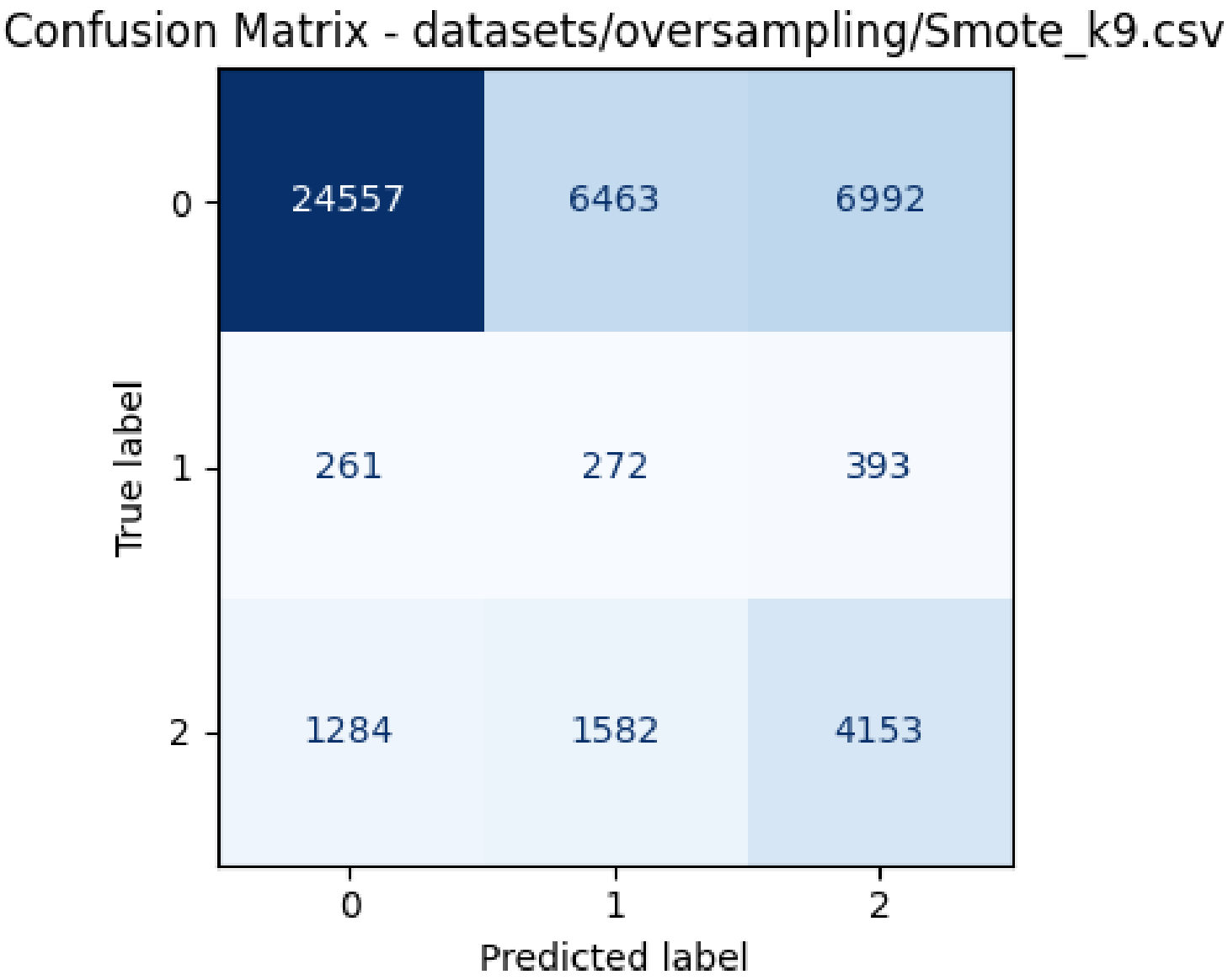

Figure 4.2: SMOTE K=9 Softmax Confusion Matrix

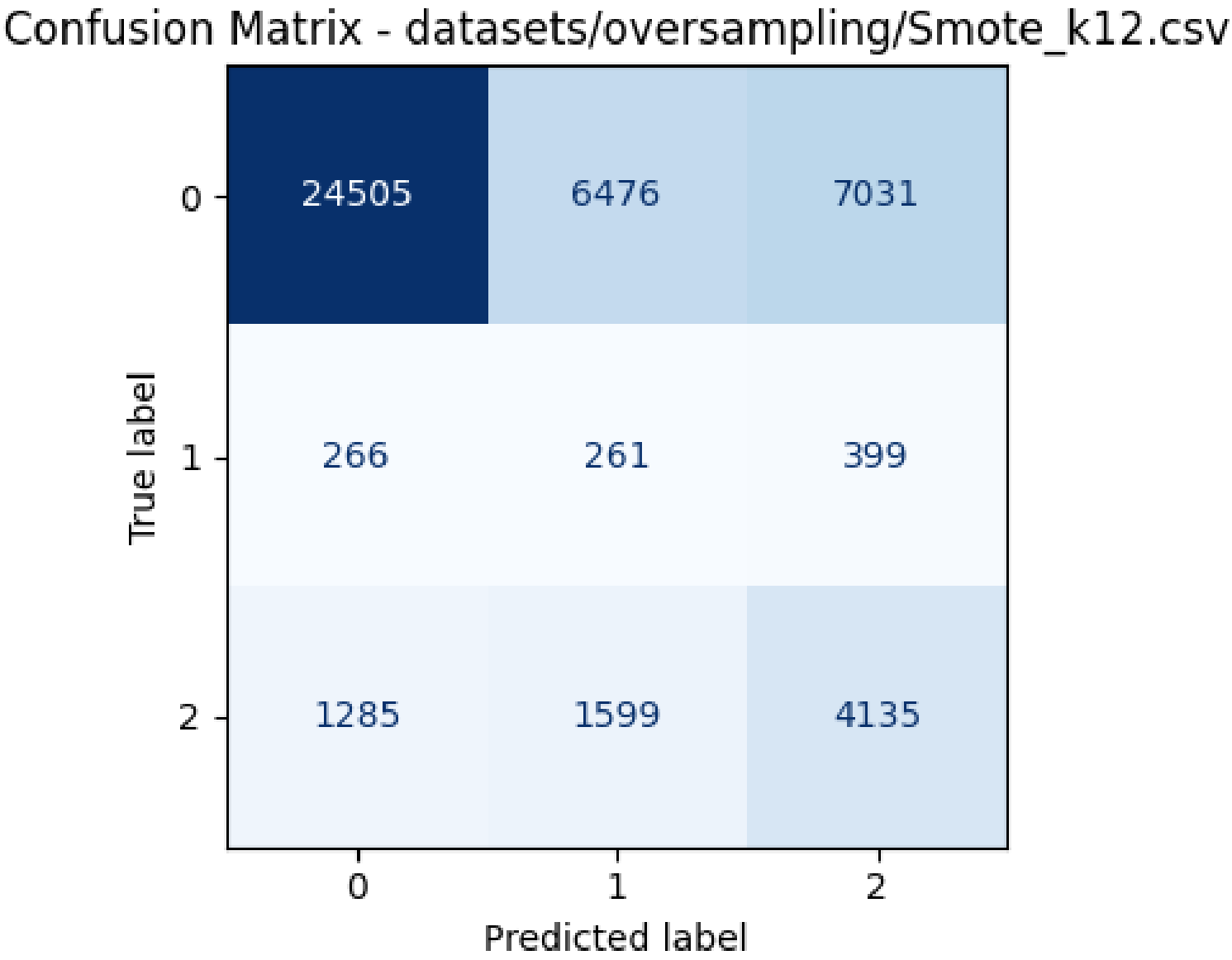

Figure 4.3: SMOTE K=12 Softmax Confusion Matrix

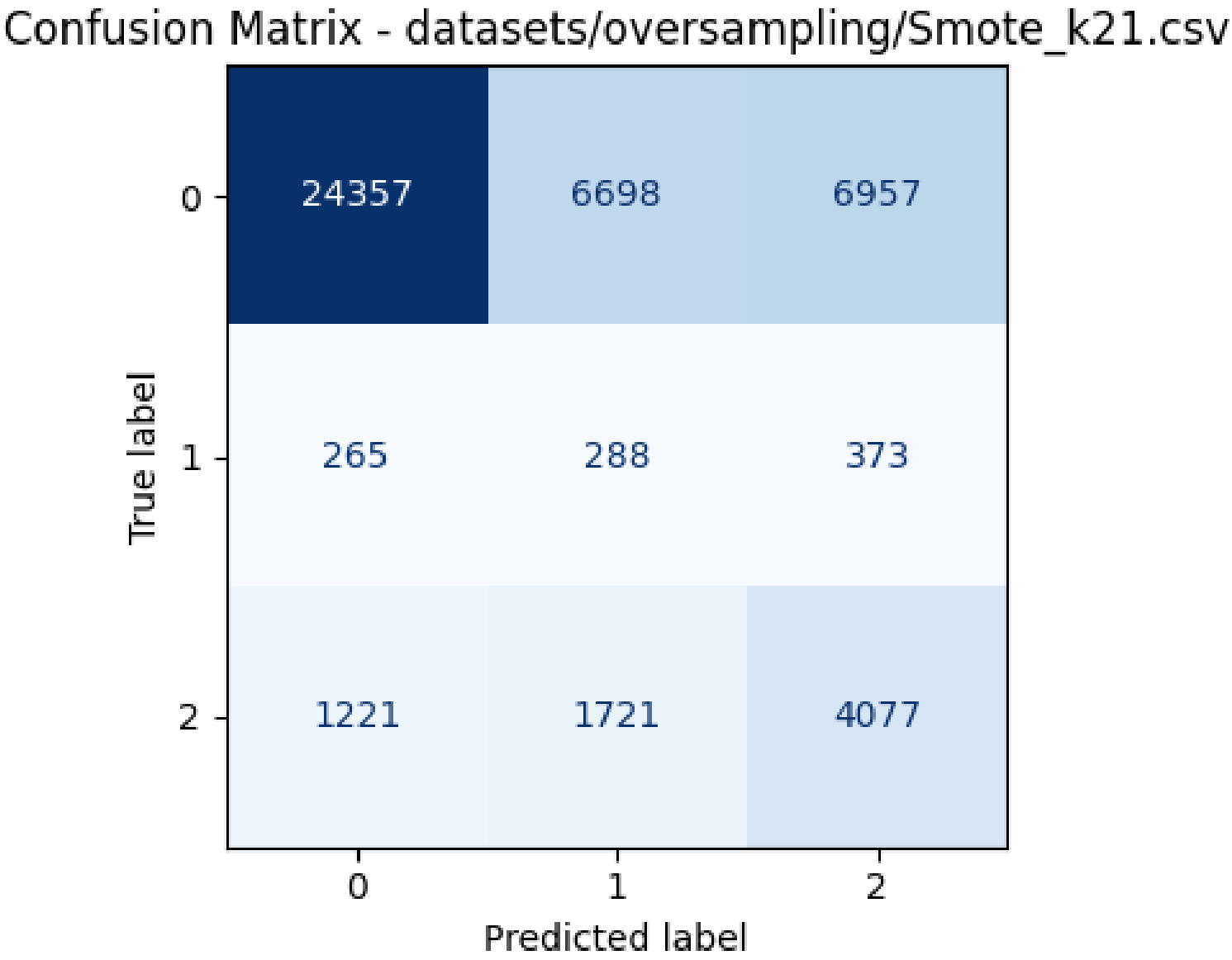

Figure 4.4: SMOTE K=12 Softmax Confusion Matrix

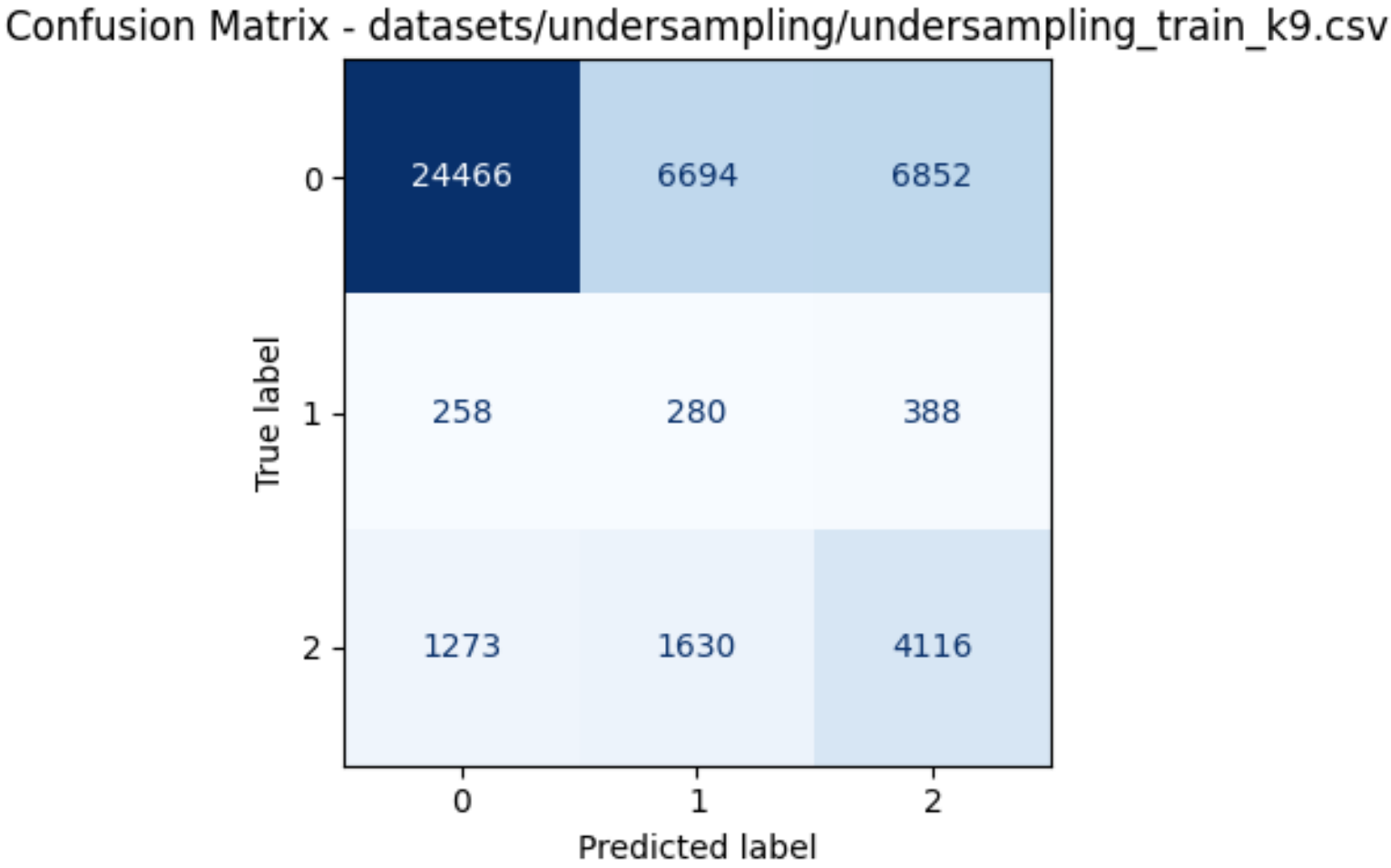

Figure 4.5: Random Under-sampling K=9 Softmax Confusion Matrix

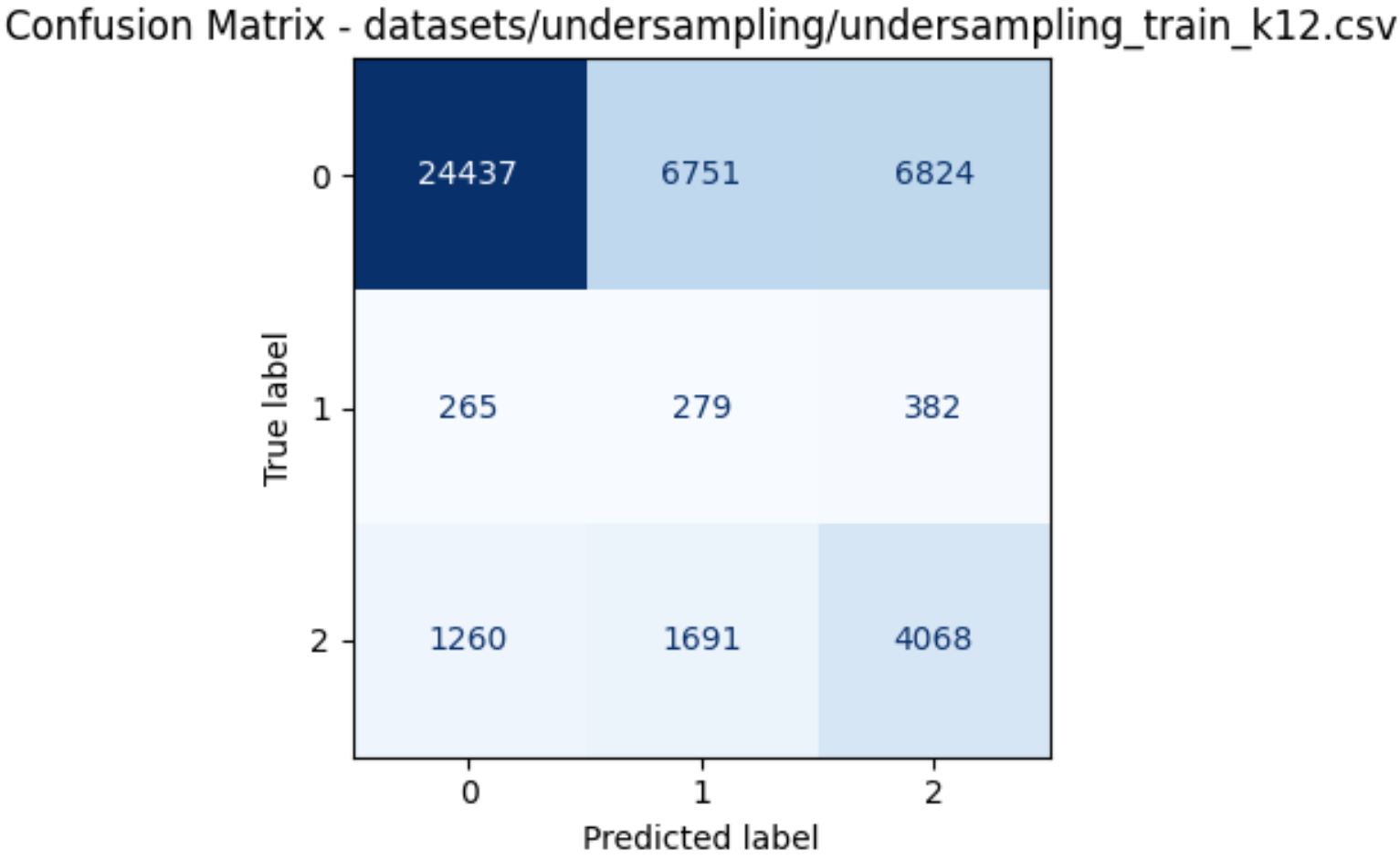

Figure 4.6: Random Under-sampling K=12 Softmax Confusion Matrix

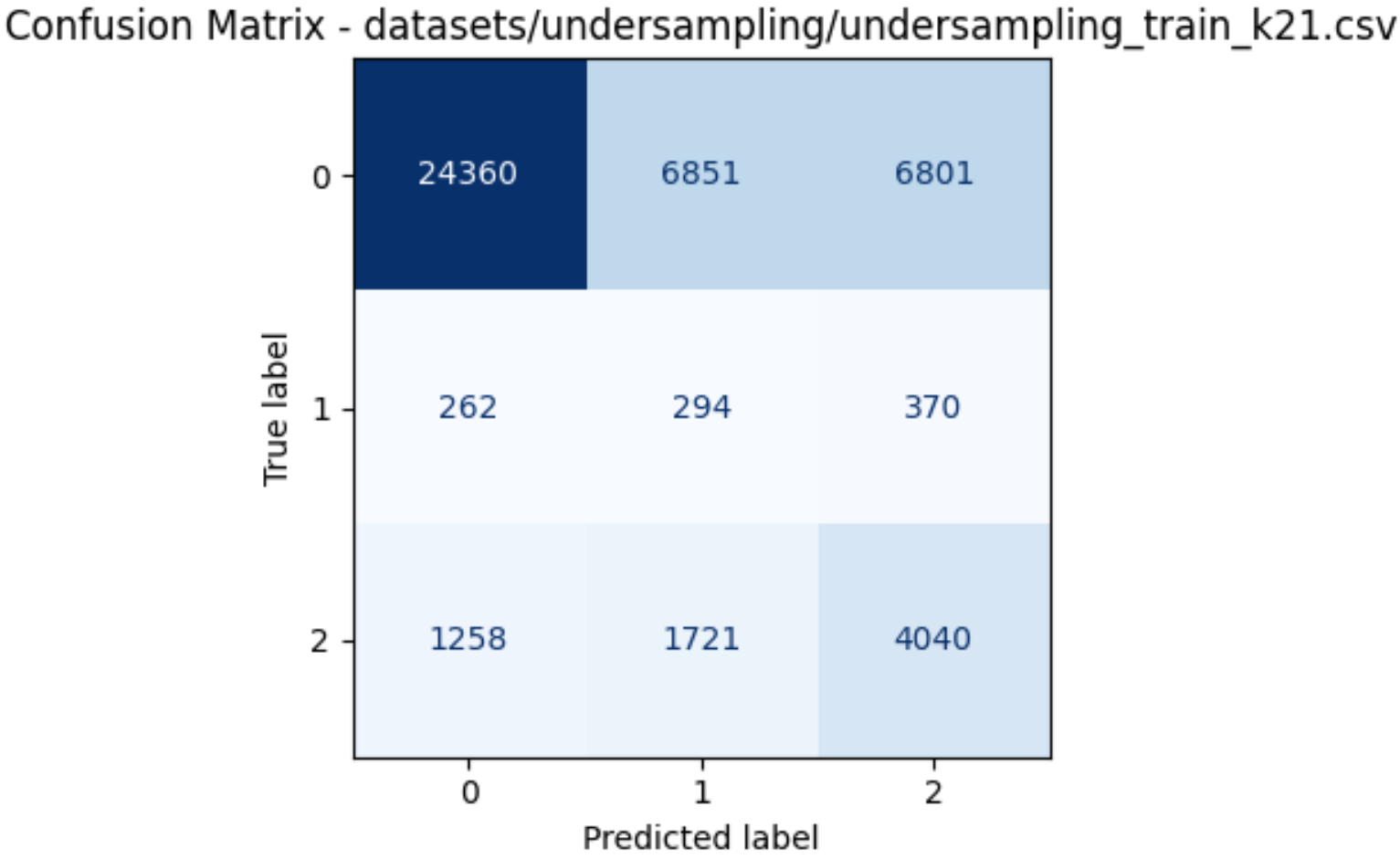

Figure 4.7: Random Under-sampling K=21 Softmax Confusion Matrix

Neural Networks

5.1 Overview

In this chapter, we will discuss the training of the Neural Networks for diabetes risk prediction.

5.2 Evaluation Metrics

Table 5.1: Neural Network Performance Metrics Across Dataset Variants

Model	Accuracy	F1 Micro	F1 Macro	F1 Weighted	Precision (W)	Recall (W)
base	0.8493	0.8493	0.3898	0.8077	0.8049	0.8493
smote_k9	0.7702	0.7702	0.4407	0.7818	0.8068	0.7702
smote_k12	0.7819	0.7819	0.4468	0.7892	0.7977	0.7819
smote_k21	0.8298	0.8298	0.4163	0.7999	0.7882	0.8298
undersample_k9	0.7048	0.7048	0.4353	0.7431	0.8199	0.7048
undersample_k12	0.6216	0.6216	0.4173	0.6929	0.8352	0.6216
undersample_k21	0.6423	0.6423	0.4326	0.7162	0.8369	0.6423

The Neural Network results show a clear trade-off between overall accuracy and minority-class balance. The baseline model achieves the highest **Accuracy** and **F1 Micro**, which indicates strong performance on the dominant classes, while SMOTE-based variants improve **F1 Macro**, suggesting better class-wise balance across all categories. In particular, `smote_k12` provides the strongest macro-level performance among the SMOTE settings, whereas under-sampling configurations generally reduce global performance despite relatively competitive weighted precision and recall values. These findings indicate that data-balancing strategy has a direct impact on whether the model prioritizes aggregate correctness or more equitable multi-class behavior.

5.3 Confusion Matrix

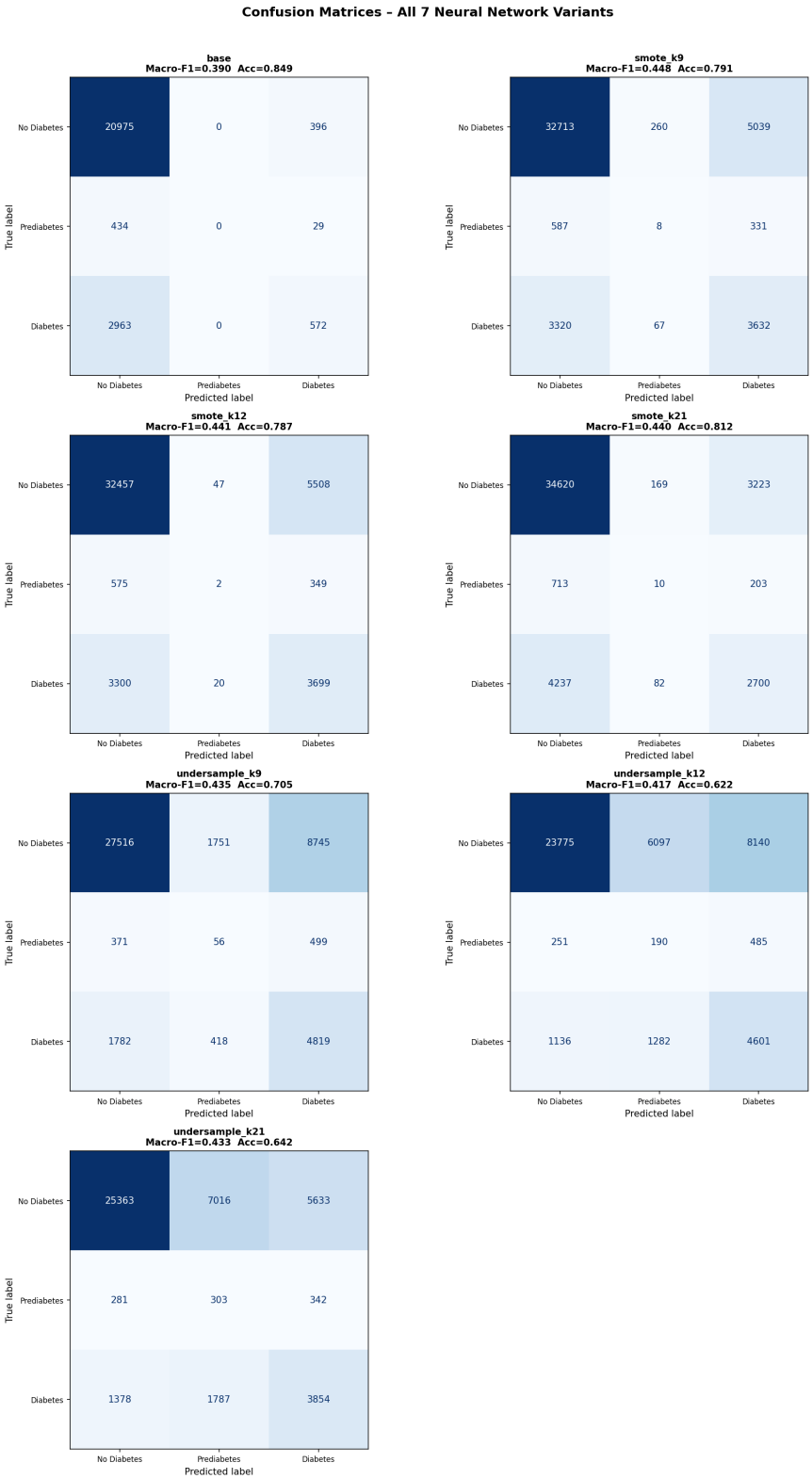

Figure 5.1: Baseline Neural Network Confusion Matrix

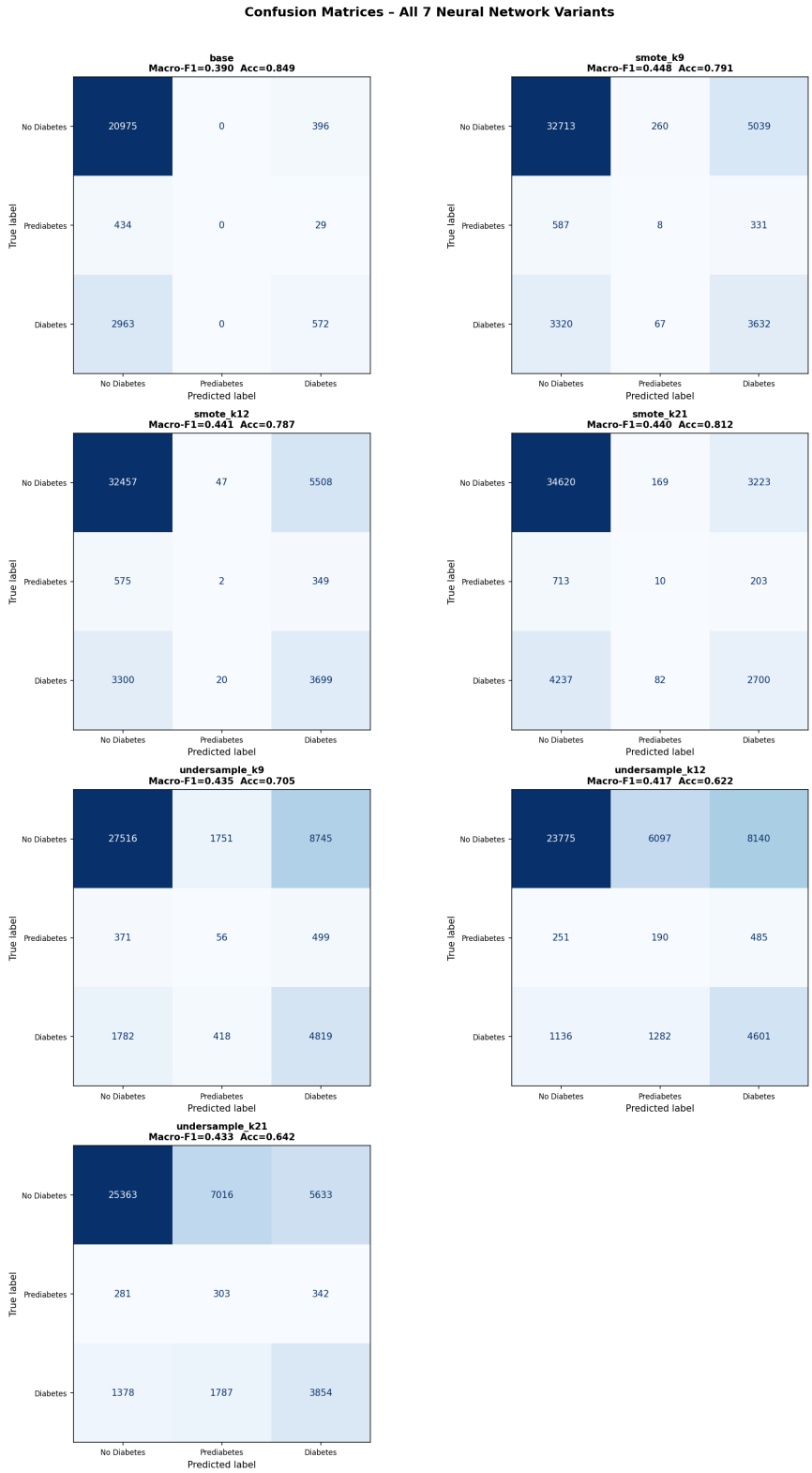

Figure 5.2: Confusion Matrices for All Neural Network Models

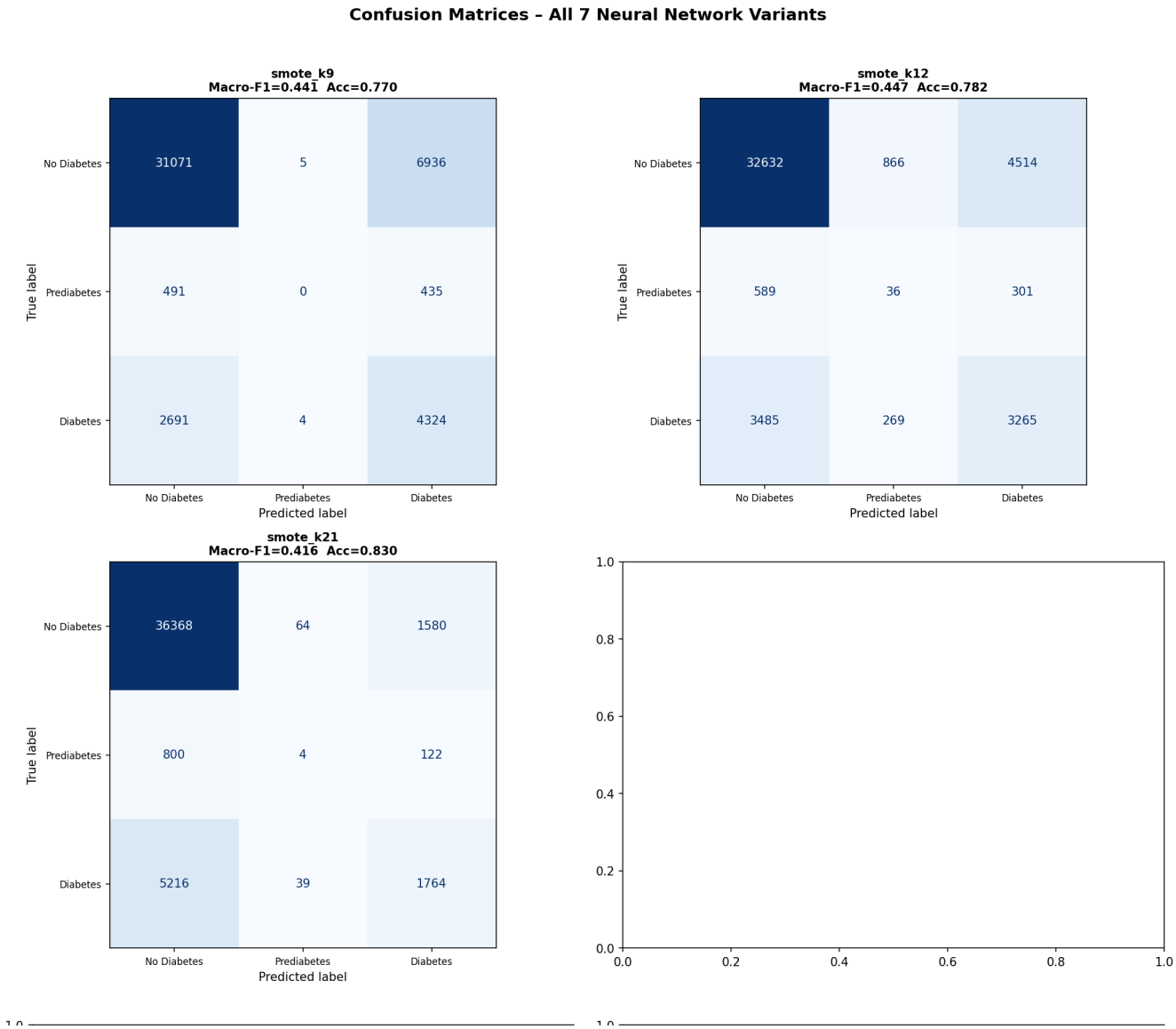

Figure 5.3: Confusion Matrices for All SMOTE Neural Network Models

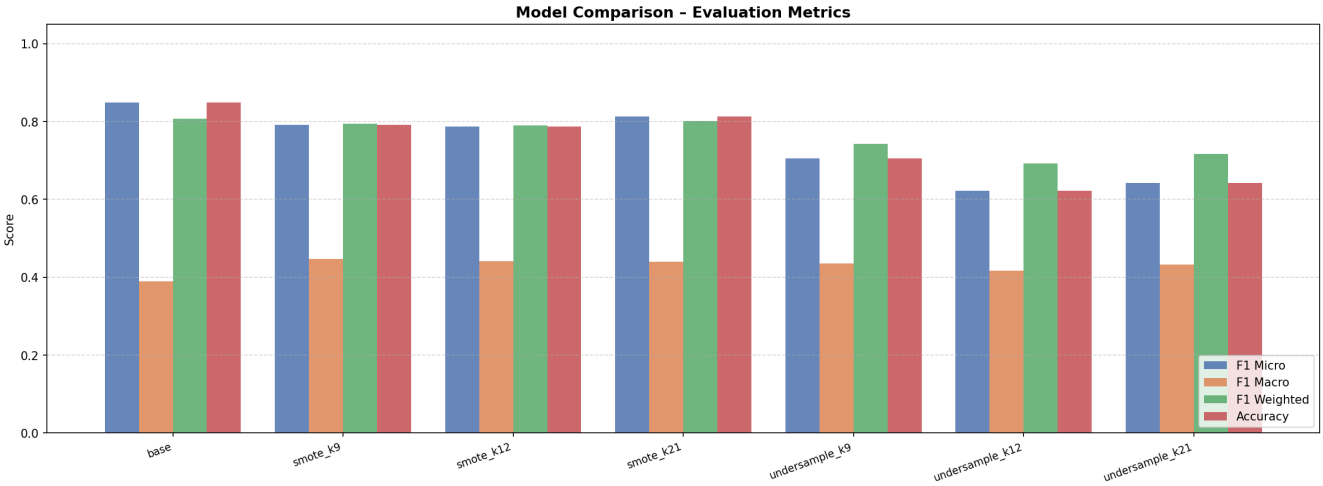

Figure 5.4: Metrics Comparison for All Neural Network Models

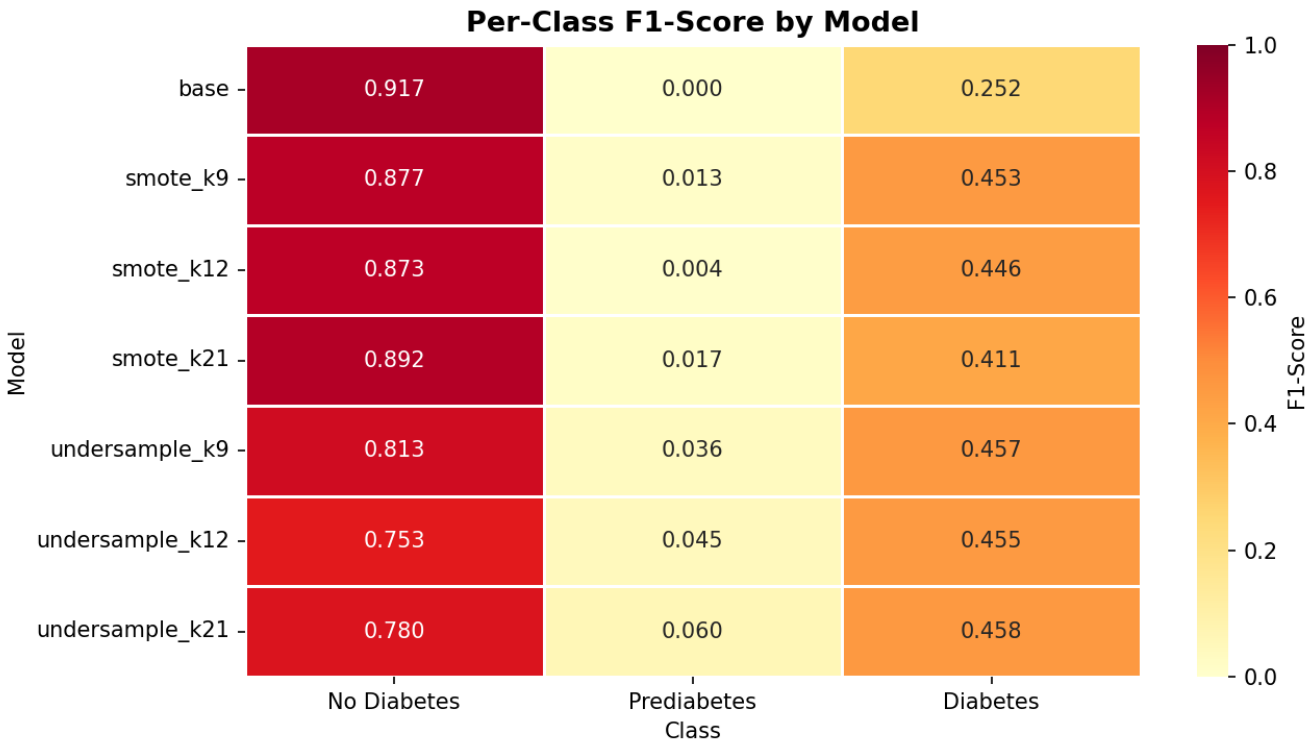

Figure 5.5: Per-Class F1-Score Heatmap for Neural Network Models

5.4 TensorBoard Visualizations

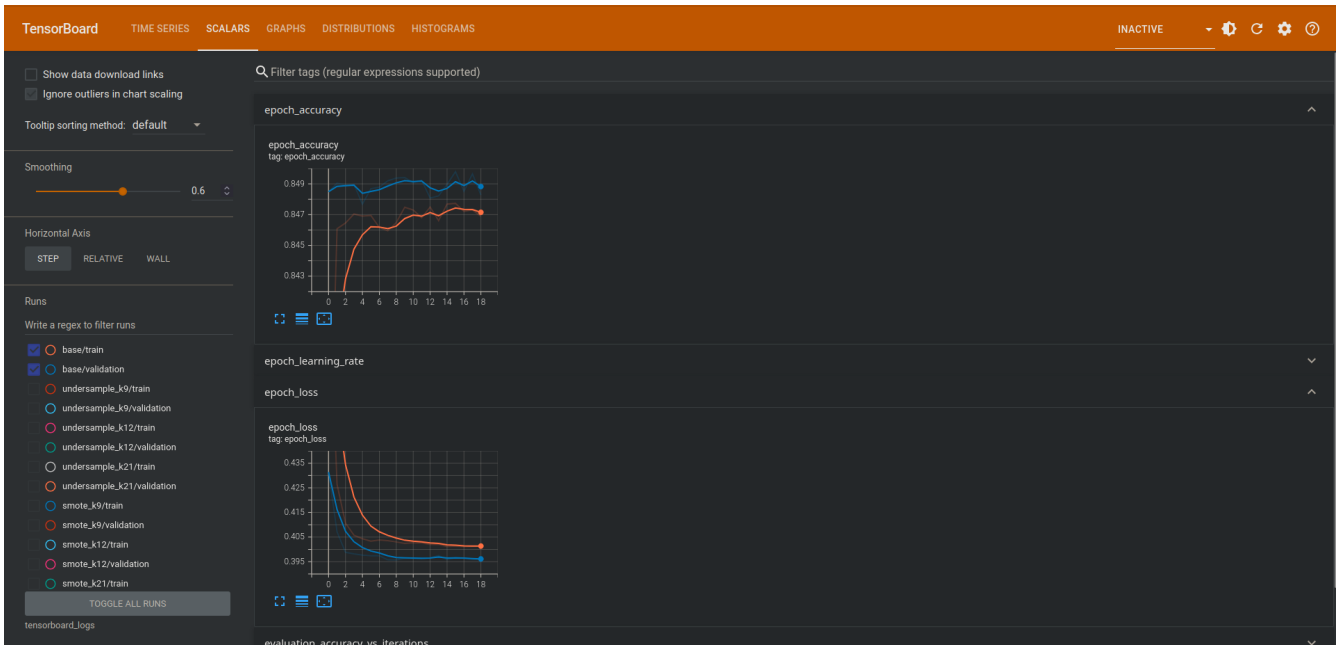

Figure 5.6: Baseline TensorBoard

Observation: The baseline run appears to have stable learning with no strong train-validation divergence, so severe overfitting is not obvious. However, relative to balanced runs, it likely underfits minority classes because optimization is dominated by the majority class.

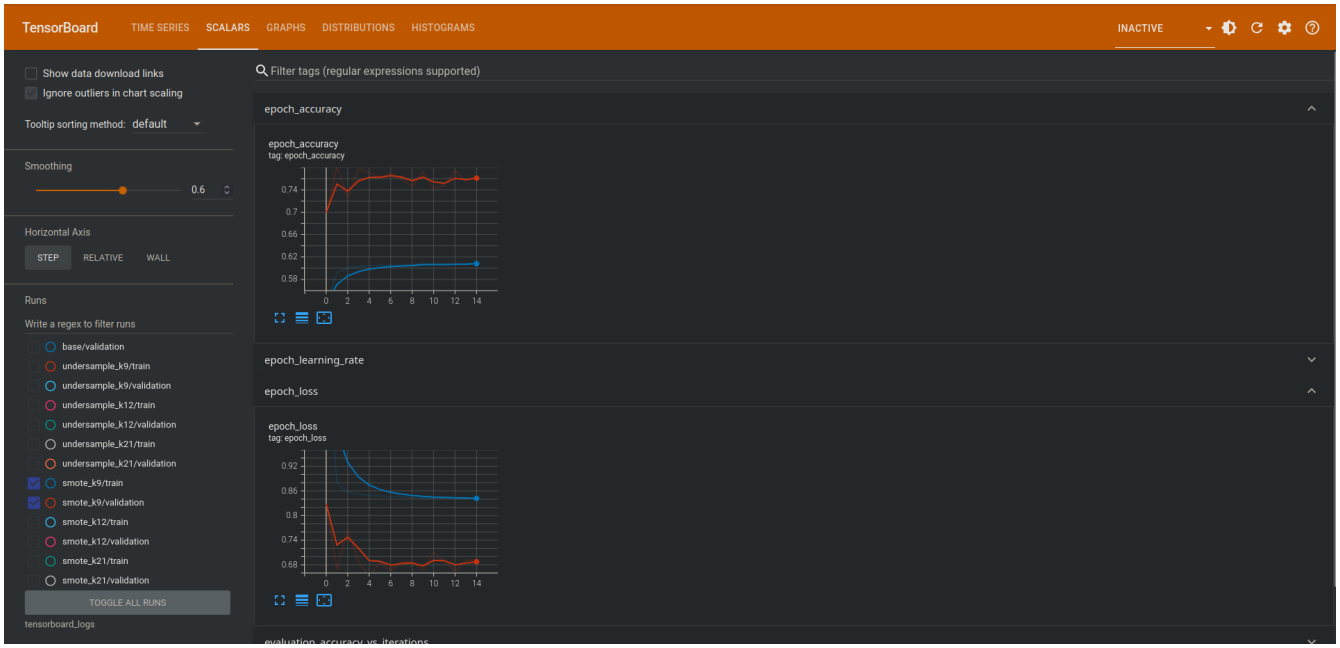

Figure 5.7: SMOTE K=9 TensorBoard

Observation: This run shows generally healthy convergence and only mild late-epoch gap behavior, suggesting limited overfitting. It looks better balanced than the baseline and does not show clear underfitting.

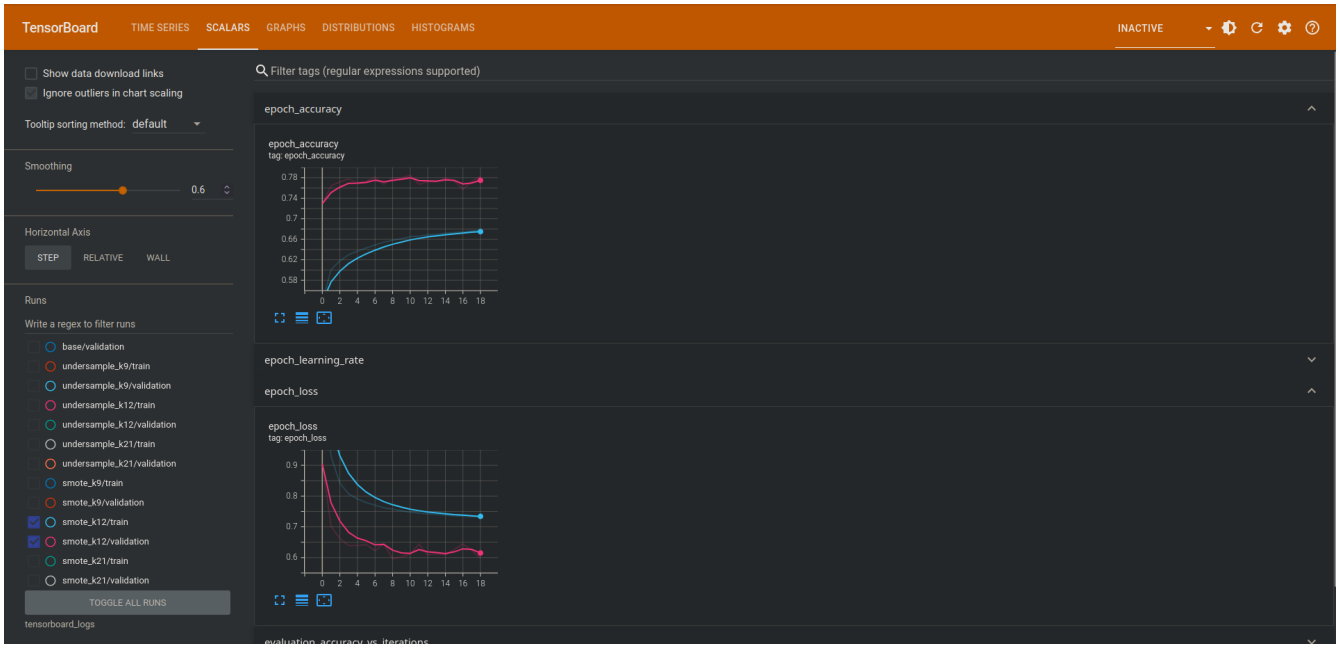

Figure 5.8: SMOTE K=12 TensorBoard

Observation: The curves are the most consistent among the SMOTE settings, with good validation tracking and no large persistent gap. This indicates the best trade-off here, with weak signs of overfitting and no meaningful underfitting.

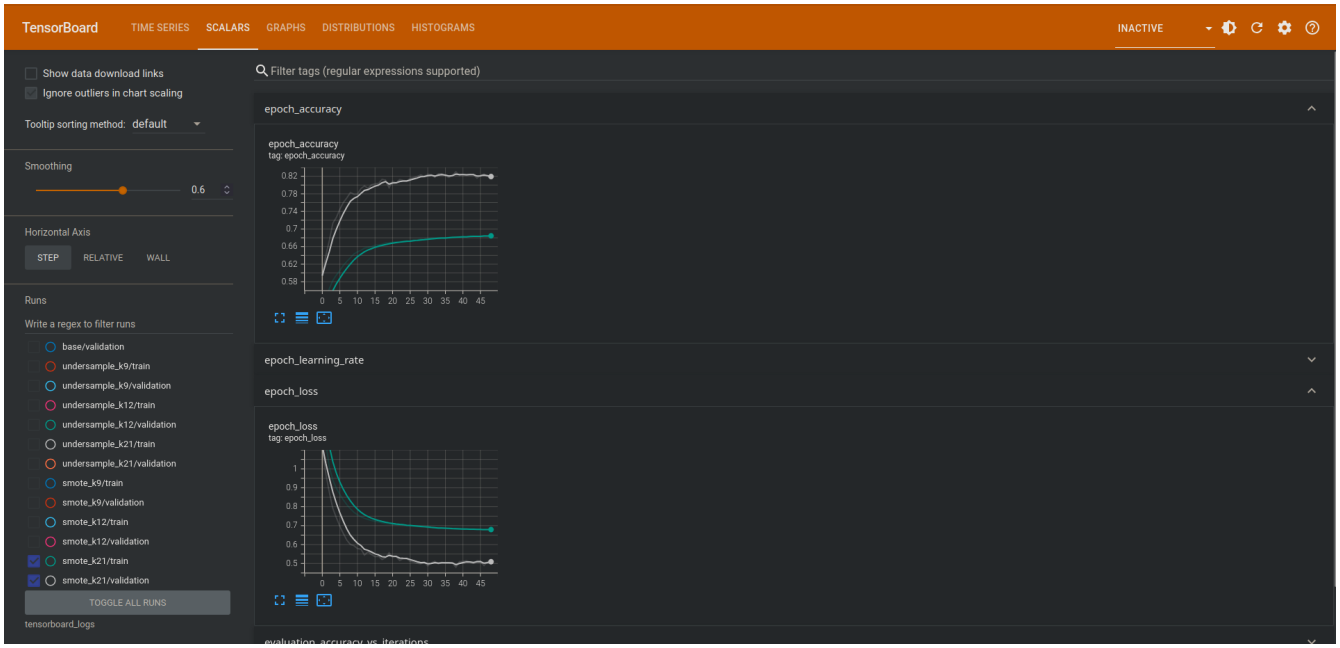

Figure 5.9: SMOTE K=21 TensorBoard

Observation: Compared with SMOTE $k = 12$, this setting shows slightly weaker generalization behavior and a modest train-validation gap. That suggests mild overfitting pressure without clear severe underfitting.

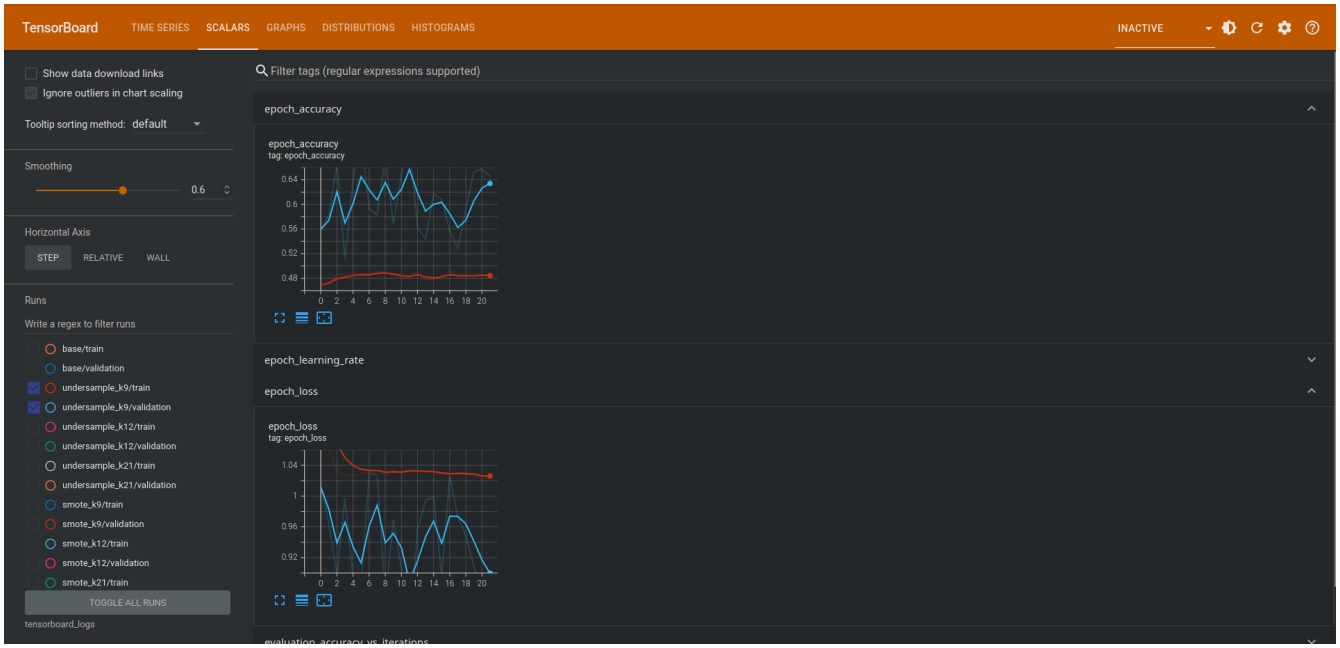

Figure 5.10: Random Under-sampling K=9 TensorBoard

Observation: Under-sampling at $k = 9$ tends to converge quickly but to a lower performance ceiling, which is a typical sign of underfitting due to reduced training diversity. Overfitting is limited because the model capacity is effectively constrained by less data.

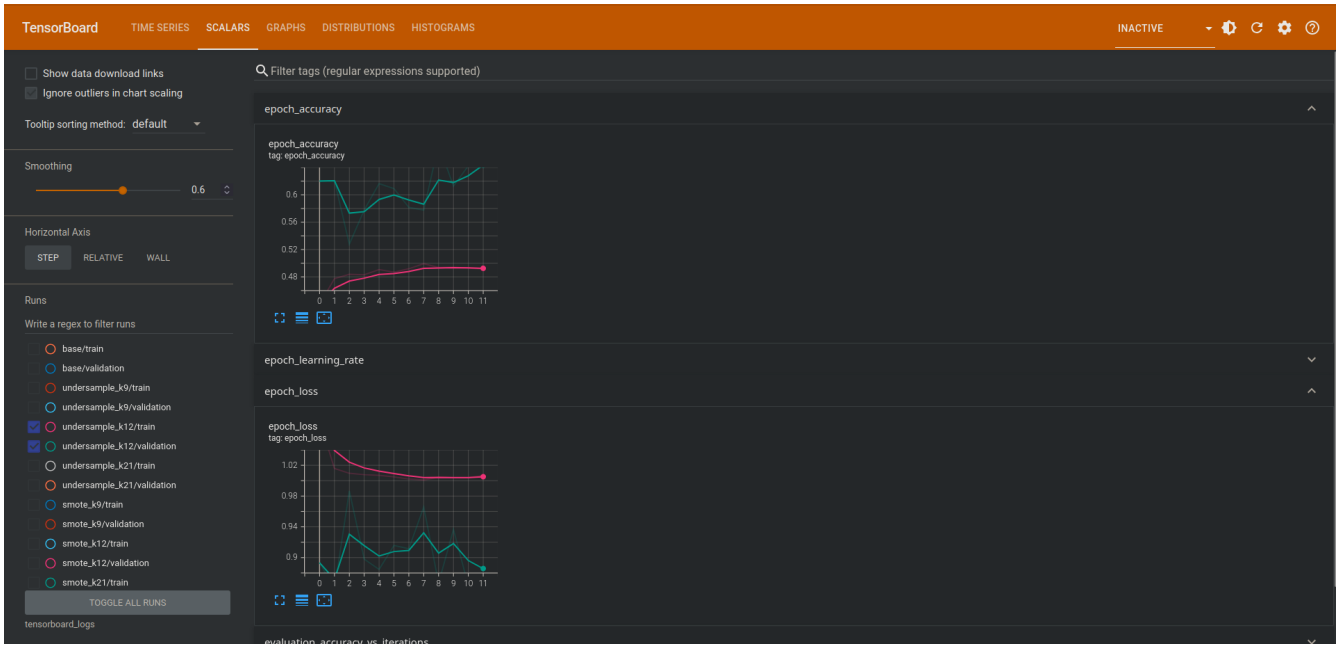

Figure 5.11: Random Under-sampling K=12 TensorBoard

Observation: This run shows the clearest underfitting tendency among the under-sampled settings, with validation improvement saturating early and weaker final performance. There is no strong evidence of classical overfitting.

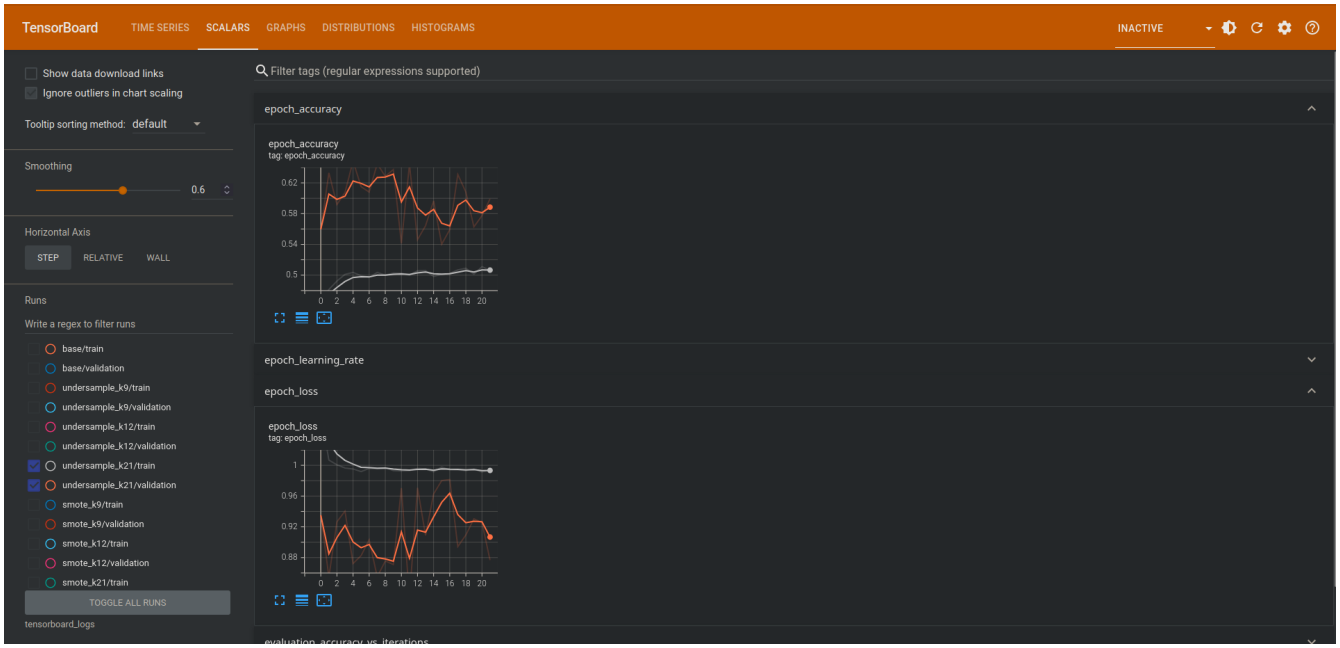

Figure 5.12: Random Under-sampling K=21 TensorBoard

Observation: The training dynamics are somewhat better than under-sampling $k = 12$, but still suggest a constrained fit relative to SMOTE configurations. Overall this indicates mild-to-moderate underfitting with limited overfitting.

Evaluation Metrics

6.1 Comparison Strategy

To compare the three model families fairly, we focus on four core metrics reported for all settings: **Accuracy**, **F1 Micro**, **F1 Macro**, and **F1 Weighted**. Accuracy and F1 Micro reflect overall correctness, while F1 Macro gives equal importance to all classes and is therefore more sensitive to minority-class performance.

6.2 Baseline Comparison

Table 6.1: Baseline (Unbalanced Data) Comparison Across Model Families

Model Family	**Accuracy**	**F1 Micro**	**F1 Macro**	**F1 Weighted**
KNN	0.8219	0.8219	0.4016	0.7993
Softmax Regression	0.8335	0.8335	0.3955	0.7931
Neural Network	0.8493	0.8493	0.3898	0.8077

The baseline results show that the **Neural Network** obtains the highest overall performance (Accuracy/F1 Micro), while **KNN** provides a slightly stronger baseline **F1 Macro**. This indicates that the NN is best at global prediction correctness on the original distribution, but KNN is marginally more balanced across classes before rebalancing.

6.3 Balanced Data Comparison (Shared Settings)

6.3.1 SMOTE Comparison

Table 6.2: SMOTE Comparison Across Model Families

Setting	Model Family	Accuracy	F1 Micro	F1 Macro	F1 Weighted
$k = 9$	KNN	0.7298	0.7298	0.4276	0.7579
$k = 9$	Softmax	0.6306	0.6306	0.4242	0.7032
$k = 9$	Neural Network	0.7702	0.7702	0.4407	0.7818
$k = 12$	KNN	0.7302	0.7302	0.4315	0.7592
$k = 12$	Softmax	0.6289	0.6289	0.4221	0.7018
$k = 12$	Neural Network	0.7819	0.7819	0.4468	0.7892
$k = 21$	Softmax	0.6250	0.6250	0.4217	0.6998
$k = 21$	Neural Network	0.8298	0.8298	0.4163	0.7999

Under SMOTE, the **Neural Network** gives the strongest Accuracy/F1 Micro in all reported settings and also achieves the best F1 Macro at $k = 9$ and $k = 12$. KNN remains competitive in macro performance at those settings, while Softmax is consistently lower across the four core metrics.

6.3.2 Under-sampling Comparison

Table 6.3: Random Under-sampling Comparison Across Model Families

Setting	Model Family	Accuracy	F1 Micro	F1 Macro	F1 Weighted
$k = 9$	KNN	0.5870	0.5870	0.4020	0.6714
$k = 9$	Softmax	0.6280	0.6280	0.4237	0.7019
$k = 9$	Neural Network	0.7048	0.7048	0.4353	0.7431
$k = 12$	KNN	0.5908	0.5908	0.4017	0.6735
$k = 12$	Softmax	0.6263	0.6263	0.4222	0.7010
$k = 12$	Neural Network	0.6216	0.6216	0.4173	0.6929
$k = 21$	Softmax	0.6244	0.6244	0.4219	0.6996
$k = 21$	Neural Network	0.6423	0.6423	0.4326	0.7162

For under-sampling, the Neural Network performs best at $k = 9$ and $k = 21$, but Softmax slightly surpasses it at $k = 12$. KNN is consistently the weakest under under-sampling in this report, suggesting that aggressive class reduction harms neighborhood-based learning more than parametric models.

6.4 Best Configurations by Objective

Table 6.4: Best Observed Configurations Across All Reported Results

Objective	Best Configuration	Value
Highest Accuracy / F1 Micro	Neural Network (base)	0.8493
Highest F1 Macro	Neural Network (smote_k12)	0.4468
Highest F1 Weighted	Neural Network (base)	0.8077

Overall, the results indicate that **Neural Networks** provide the strongest performance envelope across both global and balanced metrics. **KNN** remains a viable baseline and can achieve competitive macro performance under SMOTE, while **Softmax Regression** is stable but generally trails in top-end performance. Therefore, model selection should follow the deployment goal: if maximum global accuracy is required, the NN baseline is preferred; if improved minority-class balance is prioritized, NN with SMOTE (especially $k = 12$) is the best setting in this study.

6.5 Best Evaluation Metric for Model Selection

Given the class imbalance in the diabetes dataset, **F1 Macro** is the most appropriate metric for model selection in this context. It provides a balanced view of performance across all classes, ensuring that improvements in minority-class predictions are not overshadowed by the majority class. While Accuracy and F1 Micro can be misleading in imbalanced scenarios, F1 Macro offers a more nuanced assessment of how well the model performs across the entire class distribution, making it the best choice for selecting the most effective model for diabetes risk prediction in this study.

Discussion

7.1 Best Models

Table 7.1: Best Model per Algorithm (Selected by Highest F1 Macro)

Algorithm	Best Configuration	Accuracy	F1 Micro	F1 Macro	F1 Weighted
KNN	SMOTE ($k = 12$)	0.7302	0.7302	0.4315	0.7592
Softmax Regression	SMOTE ($k = 9$)	0.6306	0.6306	0.4242	0.7032
Neural Network	SMOTE ($k = 12$)	0.7819	0.7819	0.4468	0.7892

This summary highlights each algorithm's strongest configuration under the class-balance criterion used in this report (F1 Macro).

7.1.1 Why These Were the Best Models

The selected configuration for each algorithm is the one that achieved the highest **F1 Macro** within that algorithm's tested settings. This criterion is appropriate because the diabetes classes are imbalanced, so equal class-wise performance is more informative than accuracy alone.

For **KNN**, SMOTE ($k = 12$) is best because it yields the top macro-level balance for KNN (**F1 Macro = 0.4315**) while preserving solid overall performance (Accuracy/F1 Micro = 0.7302).

For **Softmax Regression**, SMOTE ($k = 9$) is best because it provides the strongest class-balanced Softmax result (**F1 Macro = 0.4242**) and is also the best Softmax setting by weighted F1 among the reported balanced variants.

For **Neural Networks**, SMOTE ($k = 12$) is best because it achieves the highest macro performance not only within NN (**F1 Macro = 0.4468**) but also across all reported model families and settings, indicating the strongest minority-class handling.

7.1.2 Comparison Between the Three Best Models

Comparing the three best-per-algorithm models directly:

- **Neural Network (SMOTE $k = 12$)** outperforms **KNN (SMOTE $k = 12$)** by +0.0517 Accuracy/F1 Micro, +0.0153 F1 Macro, and +0.0300 F1 Weighted.
- **Neural Network (SMOTE $k = 12$)** outperforms **Softmax (SMOTE $k = 9$)** by +0.1513 Accuracy/F1 Micro, +0.0226 F1 Macro, and +0.0860 F1 Weighted.
- **KNN (SMOTE $k = 12$)** outperforms **Softmax (SMOTE $k = 9$)** by +0.0996 Accuracy/F1 Micro, +0.0073 F1 Macro, and +0.0560 F1 Weighted.

Overall, the best NN model is the strongest in both global and balanced metrics, the best KNN model is second and remains reasonably competitive on F1 Macro, and the best Softmax model is the most conservative performer with the lowest scores among the three selected winners.

7.2 Imbalance Handling Effectiveness

Table 7.2: Imbalance-Handling Effectiveness by Algorithm (Macro-F1 Gain)

Algorithm	**Baseline**	**Best Setting**	**Best Macro-F1**	**Gain**	**Gain (%)**
KNN	0.4016	SMOTE ($k = 12$)	0.4315	+0.0299	+7.44%
Softmax Regression	0.3955	SMOTE ($k = 9$)	0.4242	+0.0287	+7.26%
Neural Network	0.3898	SMOTE ($k = 12$)	0.4468	+0.0570	+14.62%

The effectiveness table shows that class-imbalance treatment improved Macro-F1 for all three algorithms. The Neural Network achieved the largest absolute and relative gain, indicating the strongest improvement in minority-class sensitivity after balancing.

7.3 Insights and Observations

Analysis of the undersampling_train_k12 confusion matrix reveals significant misclassification patterns, particularly surrounding the minority "Prediabetes" class. Out of 926 true prediabetic cases, the model only identified 98 correctly, misclassifying 228 as "No Diabetes" and 600 as "Diabetes". Clinically, this occurs because prediabetes is a transitional state; without precise lab metrics (like A1C), the survey-based lifestyle features (BMI, HighBP) of a prediabetic patient heavily overlap with those of a diabetic patient. Mathematically, optimizing for Macro-F1 caused the Softmax decision boundaries to shift. Because the feature space for Class 1 is poorly defined and overlaps heavily with the other classes, the algorithm minimized its precision penalty by absorbing ambiguous cases into the broader "Diabetes" class. Additionally, the imbalance handling created a hypersensitive model, resulting in 9,485 healthy patients being falsely flagged as diabetic, illustrating the strict mathematical trade-off between minority class recall and majority class precision.